# Adaptive Calibration for Cross-Domain Few-Shot Learning Hyperspectral Image Classification

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***Abstract*—Coastal wetland hyperspectral image classification (HSIC) is practically constrained by scarce labeled samples. Cross-domain few-shot learning (CDFSL) provides a promising solution by enabling knowledge transfer under limited supervision. However, coastal wetlands exhibit high spectral similarity and spatial heterogeneity. Existing CDFSL methods struggle to disentangle such complex geomorphology, leading to insufficient feature discriminability. Furthermore, severe prototype estimation bias and domain shift persist as inherent challenges. To address these issues, an adaptive calibration strategy for CDFSL is proposed. Specifically, a collaborative perception feature extractor (CPFE) with a spectral–spatial adaptive differencing module (SSAD) and a dual-stream gated fusion module (DGFM) is constructed to extract the fine-grained signatures and complex spatial structures. Building upon these robust representations, we design a dynamic prototype interaction calibration (DPIC) module, which adaptively distinguishes typical samples from hard samples via the prediction confidence of query samples, thereby constructing and fusing multiple auxiliary prototypes to enable class prototypes to better approximate the latent data distribution. Finally, we develop a synergistic global domain alignment (SGDA) strategy that jointly aligns macroscopic spatial distributions and spectral baselines while preserving class-discriminative structures. Extensive experiments on three self-constructed hyperspectral image (HSI) datasets and one public dataset demonstrate the superiority of the proposed CDFSL method for HSIC via adaptive calibration (ACFSL).**

***Index Terms*—Adaptive calibration, coastal wetland, cross-domain, few-shot learning (FSL), hyperspectral image (HSI) classification.**

Received 25 April 2026; revised 7 July 2026 and 27 August 2026; accepted 28 September 2026. Date of publication 1 October 2026; date of current version 15 October 2026. This work was supported in part by Shandong Provincial Natural Science Foundation under Grant ZR2024MF037, in part by the Postdoctoral Innovation Program of Shandong Province under Project SDCX-ZG-202502039, in part by the Postdoctoral Program of Qingdao under Grant QDBSH20250102053, and in part by the Fundamental Research Funds for the Central Universities under Grant 25CX06046A. *(Corresponding author: Fangming Guo.)*

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Digital Object Identifier 10.1109/TGRS.2026.3739339

## NOMENCLATURE

| | |
|---|---|
| *SSAD* | |
| $x$ | Input feature tensor. |
| $\gamma_c/\gamma_{h_w}$ | Attention weights and bias parameters. |
| *DGFM* | |
| $x$ | Input feature is reshaped into a sequence representation. |
| $P_h/P_u$ | Positional embeddings. |
| $c(\cdot)$ | Channel-wise attention. |
| $\mathcal{M}(\cdot)$ | Spatial-wise attention. |
| *DPIC* | |
| $\ell(\cdot)$ | Logit of query sample prediction. |
| $\bar{m}_k$ | Class-level difficulty indicator. |
| $\overline{\boldsymbol{p}}_k^N/\overline{\boldsymbol{p}}_k^H$ | Typical prototype and the hard prototype of the $k$th class. |
| $p_k^*$ | Calibrated prototype features. |
| *SGDA* | |
| $\mathcal{F}_{\text{map}}$ | Shallow spatial feature maps. |
| $W_{dw}$ | Depthwise convolution kernel. |
| $\mathcal{F}$ | Reproducing kernel Hilbert space. |
| $\mathbb{E}_z[4]$ | Mathematical expectation of probability distributions. |
| $\gamma$ | Hyperparameter for scheduling rate. |
| $\tilde{\bar{\boldsymbol{\nu}}}_{d,\text{low}}$ | Empirical mean vector. |

## I. INTRODUCTION

HYPERSPECTRAL images (HSIs) capture fine-grained spectral traits of ground objects over hundreds of contiguous bands and preserve 2-D spatial structural information, thus exhibiting unique advantages in ecological monitoring, precision agriculture, and resource investigation [1], [2], [3], [4], [5]. As a core task in hyperspectral interpretation, HSI classification (HSIC) is crucial for accurate vegetation identification and biomass inversion in coastal wetlands [6], [7], [8]. Deep learning has driven convolutional neural networks (CNNs) and their variants [9], [10], [11] to achieve remarkable progress in HSIC. In recent years, the introduction of graph neural networks (GNNs) [12], [13], Transformers [14], [15], and the novel sequence modeling architecture Mamba [16], [17], [18] has further enhanced the representational capability of models. More recently, integrating graph topologies with generative transformer architectures has demonstrated significant potential [19]. Nonetheless, these methods generally rely on abundant and high-quality labeled data. Fig. 1(a) shows a

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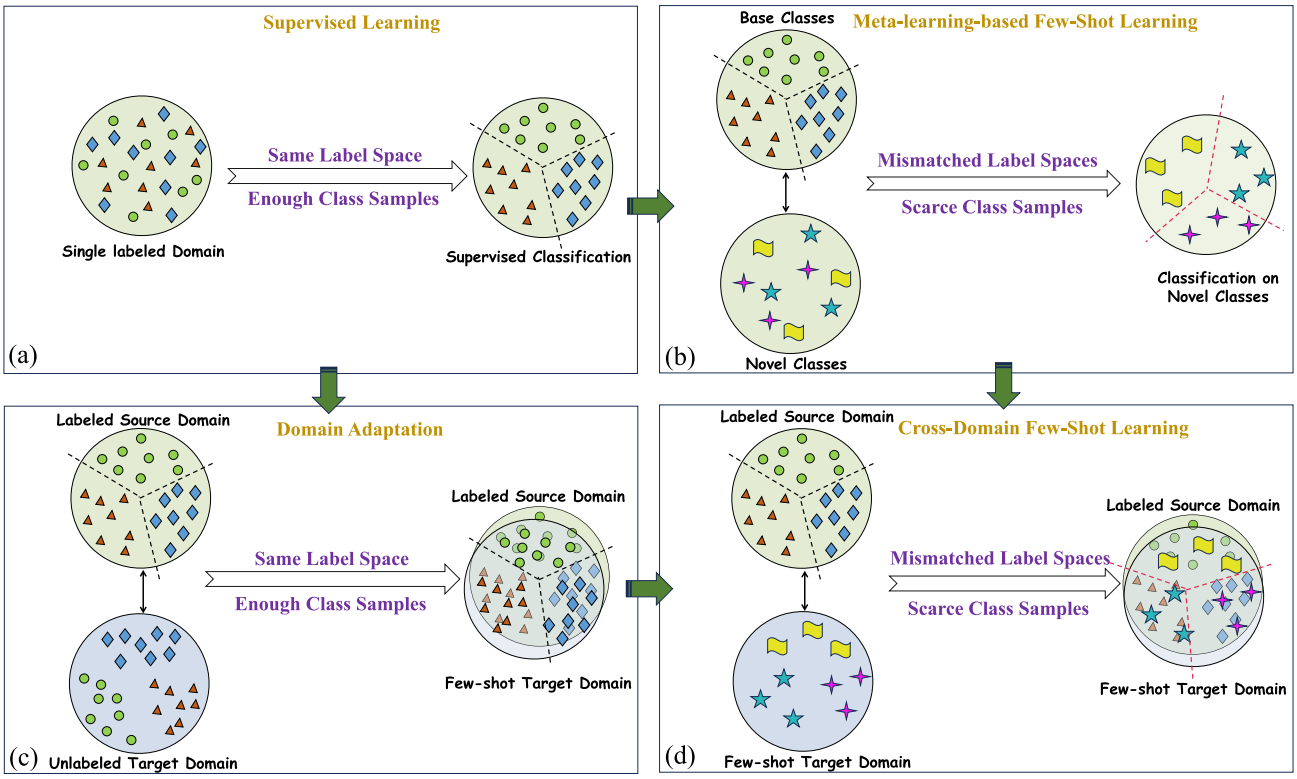



Fig. 1. Comparison of four learning paradigms. (a) Supervised learning. (b) Meta-learning-based FSL. (c) DA. (d) CDFSL.

conventional supervised learning paradigm in a single domain setting with sufficient annotations, where stable interclass decision boundaries are learned from a large number of labeled samples. However, the Yellow River Delta coastal wetland ecosystem is highly dynamic and complex [20]. Field sampling and label annotation in this region are labor-intensive and costly, making it difficult to obtain sufficient labeled samples.

To reduce reliance on extensive annotations, some semi-supervised methods adopt kernel-based normalized spectral clustering to extract discriminative information from unlabeled structures [21]. To further alleviate the scarcity of labeled data, few-shot learning (FSL) has emerged as a promising paradigm [22]. FSL is commonly formulated under a meta-learning framework [23], [24]. Episodes are constructed from datasets with sufficient labeled classes; the model learns class representations from the support set and is updated via back-propagation to minimize classification loss on the query set, enabling rapid adaptation from few samples. As illustrated in Fig. 1(b), standard FSL often implicitly assumes that training and testing data are identically distributed, and it does not explicitly address the domain shift. Consequently, distribution shifts between the source domain (SD) and target domain (TD) can lead to marked performance degradation [25], [26], [27].

To enhance the generalization ability of models in new HSIC scenarios, researchers have further investigated strategies such as pretraining-based transfer learning [28] and domain adaptation (DA) [29]. For instance, Su et al. [30] developed advanced graph transfer networks to bridge structural discrepancies and align representations across domains. Fig. 1(c) depicts the classical unsupervised DA setting. The objective is to achieve class alignment by minimizing interdomain distribution discrepancy, thereby enabling accurate classification. Specifically, Huang et al. [31] proposed an adversarial DA approach to facilitate DA for wetland mapping. Li et al. [32] mitigated the pronounced spectral shift by introducing a cyclic self-training mechanism. Ahmad et al. [33] proposed a transformer-driven active transfer learning framework for cross-HSI classification via informative sample selection. Zhang et al. [34] developed a bi-directional DA framework with a transformer architecture to enhance cross-domain HSI classification. To tackle the open-set cross-scene classification scenario with unknown classes in the TD, Liu et al. [35] developed a single-stage open-set DA method via decoupled alignment, which independently mitigates spectral and spatial distribution discrepancies between domains. Although these methods improve the performance to some extent, their classical settings usually require SD and TD to share an identical label space.

Recently, several studies [24], [36] have investigated distribution discrepancies within the meta-learning paradigm and identified cross-domain FSL (CDFSL) as an emerging research direction. As illustrated in Fig. 1(d), CDFSL aims to address both label scarcity and domain shift, requiring models to possess rapid adaptation capabilities that enable them to generalize across disparate data distributions and adapt to novel classes in TD. For example, Li et al. [27] introduced adversarial learning to mitigate distribution shift in FSL. Zhang et al. [37] leveraged graph information aggregation to align SD and TD data from both geometric and statistical perspectives. Wang et al. [38] characterized interclass discrepancies via class-level attention and assigned alignment weights accordingly. Although these studies have achieved

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certain advances, they still face several critical challenges in heterogeneous coastal wetland scenarios. First, discrepancies in sensor responses and inconsistencies in spectral dimensionality induce a significant heterogeneity gap [39], whereas existing feature extractors are often overly simplistic and struggle to jointly maintain spectral discriminability and spatial structure modeling. Second, under few-shot conditions, class prototypes derived from the mean of support set features tend to deviate from the true data distribution, undermining the stability of metric-based classification. Furthermore, most existing methods rely on single-granularity alignment, which often yields suboptimal domain invariance and may distort class discriminative structures.

To address the above issues, a novel transductive CDFSL method for HSIC via adaptive calibration, termed ACFSL, is proposed in this article. Specifically, we design a collaborative perception feature extractor (CPFE) that first generates initial spectral–spatial representations by 3-D convolutions; a spectral–spatial adaptive differencing module (SSAD) is then introduced to dynamically recalibrate key spectral channels and salient spatial locations, and a dual-stream gated fusion module (DGFM) adaptively fuses a Mamba branch for long-range spectral dependency modeling with an attention branch for global spatial context modeling, yielding robust spectral–spatial embeddings. For FSL, we develop a dynamic prototype interaction calibration (DPIC) mechanism, which leverages the predictive confidence of support and query samples to adaptively split typical and hard samples, construct auxiliary prototypes, and generate calibrated class prototypes via nonlinear fusion to reduce prototype bias. To mitigate domain shift, we propose a synergistic global domain alignment (SGDA) strategy at two complementary levels, including spatial distribution alignment and global spectral response alignment. The main contributions of this work are summarized as follows.

1) An ACFSL is explored for coastal wetland classification. Different from existing works, the ACFSL adaptively recalibrates feature responses to handle the high spectral similarity and spatial heterogeneity of wetland scenes, providing high-quality embeddings that facilitate more stable prototype calibration and effective domain alignment.
2) We design a CPFE, where SSAD amplifies subtle interclass variations by recalibrating informative spectral channels and spatial regions, while DGFM couples long-range dependency modeling with global spatial context to capture the extended semantic connectivity of slender wetland vegetation patches.
3) We present a DPIC mechanism that constructs auxiliary prototypes from query predictions and calibrates class prototypes via nonlinear mapping with difficulty-aware weighting, alleviating prototype bias caused by severe boundary ambiguity in wetland ecological transition zones.
4) We develop an SGDA strategy that jointly aligns SD and TD feature distributions from macroscopic spatial and holistic spectral perspectives, improving transferability under domain shift while preserving class separability.

The remainder of this article is organized as follows. Section II provides a comprehensive review of related work. Section III describes the proposed ACFSL in detail. Section IV reports and analyzes the experimental results. Finally, Section V concludes this article.

## II. Related Work

### A. Cross-Domain FSL

Traditional FSL typically assumes that the meta-training and meta-testing data are independent and identically distributed or, at least, drawn from closely related distributions. In contrast, CDFSL establishes a scenario that is more aligned with practical applications, where the SD and TD data are derived from heterogeneous distributions. Specifically, Cheng et al. [40] propose causal meta-transfer learning, which separates causal and noncausal factors using a mask Transformer and causal constraints to improve cross-domain generalization. Feng et al. [41] present a decoupled knowledge distillation method that decomposes knowledge interaction and refines discriminative features, alleviating spectral redundancy and domain shift. Ye et al. [42] construct positive–negative sample pairs via graph convolution and contrastive learning, aligning domain distributions and suppressing negative transfer. Yu et al. [43] perform feature reconstruction and prototype calibration, achieving lightweight domain alignment through Gaussian mixture models and optimal transport. This article focuses on CDFSL for HSIC.

CDFSL typically adopts an episode-based meta-training paradigm, in which both SD and TD data are organized into independent task episodes. Each task $\mathcal{T}$ consists of a support set $\mathcal{S}$ and a query set $\mathcal{Q}$. $\mathcal{S}$ contains a few labeled samples for intraepisode learning and adaptation, while $\mathcal{Q}$ is used to evaluate the model's performance in that episode. A task with $C$ classes and $K$ samples per class in $\mathcal{S}$ is defined as a $C$-way $K$-shot classification task. During training, all samples in SD tasks $\mathcal{T}_S$ are labeled. The TD task set $\mathcal{T}_T$ is further divided into a training split $\mathcal{T}_T^{tr}$ containing a small number of labeled samples and a test split $\mathcal{T}_T^{te}$ for final evaluation. During optimization, the model performs fast adaptation in each episode based on the $\mathcal{S}$ and updates its parameters by minimizing the prediction loss on the corresponding $\mathcal{Q}$; this process is iterated until convergence [25]. The training procedure for TD episodes follows the same routine, but the task distribution differs from that of the SD, thereby requiring the model to possess domain generalization capability. Under this paradigm, CDFSL methods for HSIC typically perform task inference based on prototypical metric learning. Class prototypes are constructed from the $\mathcal{S}$ features, and corresponding logits are then computed based on the distances between each $\mathcal{Q}$ sample and the class prototypes, such that the shared feature extractor is updated under supervision of $\mathcal{Q}$ labels. It should be emphasized that CDFSL does not require SD and TD to share an identical label space.

### B. Adaptive Calibration for FSL

The core idea of adaptive calibration is to introduce learnable and adjustable calibration modules or weighting mechanisms at key stages of meta-learning. Wang et al. [44]

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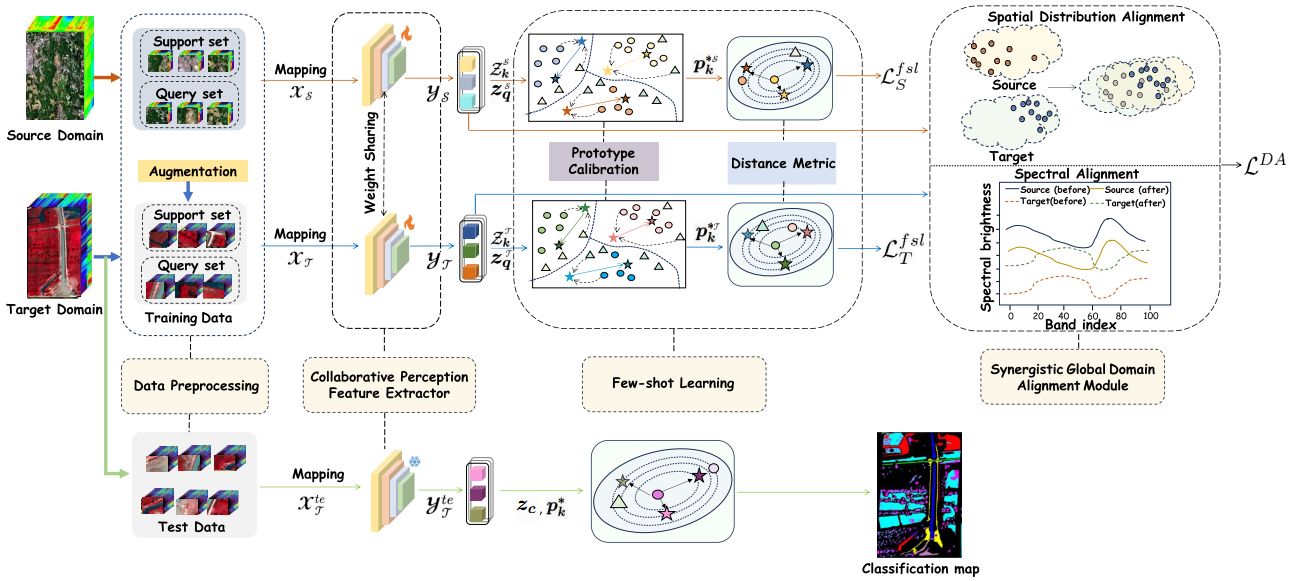



Fig. 2. Pipeline of the proposed ACFSL. The SD and TD data are divided into support and query sets, where data augmentation is applied to the limited labeled samples in the TD. A CPFE with spectral–spatial mapping is employed to learn discriminative representations. Prototype calibration and distance-based metric learning are conducted for few-shot classification, producing $\mathcal{L}_S^{fsl}$ and $\mathcal{L}_T^{fsl}$. Meanwhile, an SGDA module is introduced to reduce cross-domain distribution discrepancy, forming the domain alignment loss $\mathcal{L}^{DA}$. "Spark" represents that the parameters are trainable, while "snowflake" represents that the parameters are frozen.

fused weighted base class and novel class prototypes to calibrate novel class decision boundaries. Liu et al. [45] corrected initial prototype bias via label propagation and feature translation, considering both interclass and intraclass perspectives. Han et al. [46] employed a Gaussian mixture distribution and bipartite matching to perform posterior calibration of prototype-based decision boundaries for large language models. In CDFSL, Guo et al. [47] adaptively adjusted task-specific parameter scales and adaptation strength based on task difficulty to boost transfer efficiency. CDCNet [48] dynamically adjusts the shared feature space between SD and TD through a lightweight adaptive representation module. Jia et al. [49] enhanced robust and discriminative features via adaptive channel-spatial attention modulation. Despite these advances, adaptive calibration has yet to be systematically integrated into HSIC. Inspired by these methods, we thus design a holistic adaptive calibration mechanism. The feature extractor is equipped with SSAD and DGFM to dynamically recalibrate spectral–spatial channels and the dual path representations, while the DPIC module adaptively calibrates class prototypes to better capture intraclass structure and interclass boundaries. Together with a decoupled synergistic global alignment strategy, this framework enables the model to adapt flexibly to the distribution discrepancy between SD and TD.

## III. Proposed Method

The overall architecture of the proposed ACFSL pipeline is illustrated in Fig. 2. In preprocessing, SD and TD data are partitioned into support sets and query sets, respectively, and data augmentation is applied to the scarce labeled TD samples. Subsequently, the framework maps SD and TD hyperspectral patches into a shared spectral–spatial embedding space. The

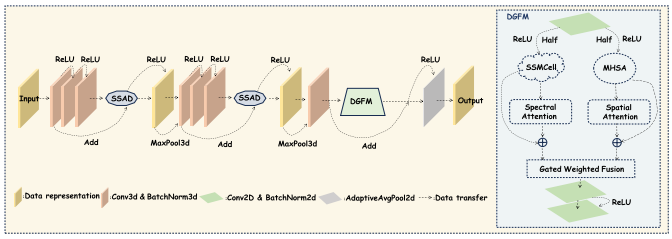



Fig. 3. Overall architecture of the CPFE.

FSL loss $\mathcal{L}^{fsl}$ is computed using a Euclidean distance-based metric classifier. An SGDA module is employed to form the DA loss $\mathcal{L}^{DA}$. The nomenclature presents the important symbols used in each module and their corresponding descriptions.

### A. Collaborative Perception Feature Extractor

The architecture of the CPFE is shown in Fig. 3, and its parameter settings are kept consistent with DCFSL [27]. A 3-D convolution backbone with residual learning is first employed to jointly extract spectral–spatial features. To further boost feature discriminability and refine representational quality, the SSAD is integrated into the backbone to perform adaptive feature refinement. The refined features are then fed into the proposed DGFM. The detailed designs of SSAD and DGFM are presented below.

a) *Spectral–spatial adaptive differencing module:* Due to the complex land cover composition of coastal wetlands, different classes often exhibit high spectral similarity and significant spatial heterogeneity, and discriminative information is usually concentrated in local spectral band variations and local spatial structural differences [50]. Targeting this characteristic, the proposed SSAD

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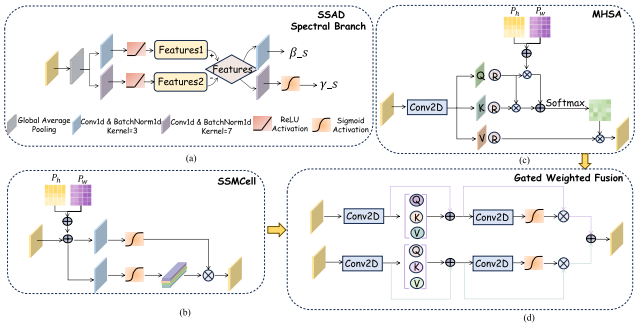



Fig. 4. Illustration of the SSAD and DGFM. (a) Spectral branch adaptive differencing. (b) SSGCell. (c) MHSA. (d) Gated feature fusion.

employs a dual-branch architecture to further conduct spectral–spatial calibration on features, aiming to amplify feature responses at informative spectral bands and salient spatial locations.

Given an input feature tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{B\times C\times S\times H\times W}$, where $B$, $C$, $S$, $H$, and $W$ denote the batch size, number of channels, number of spectral bands, spatial height, and spatial width, respectively. We take the spectral branch as an example, as shown in Fig. 4(a). This branch aims to characterize the importance of different spectral bands. First, global average pooling is applied over the spatial dimensions $H$ and $W$ to compress spatial information, yielding a spectral feature vector as

$$\boldsymbol{\mathcal{F}}_{\text{spec}} = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W}\boldsymbol{\mathcal{X}}_{[:,:,:,i,j]} \in \mathbb{R}^{B\times C\times S}. \quad (1)$$

Next, two parallel 1-D convolutional branches are employed to capture multiscale spectral context. The small kernel can capture more detailed information, so a small kernel $K_s$ ($K_s = 3$ in our experiments) is used to capture local spectral dependencies. The large kernel incorporates more background information, so a larger kernel $K_l = 2K_s + 1$ ($K_l = 7$ in our experiments) models longer range spectral correlations. Their responses are subtracted to obtain the difference, which explicitly extracts subtle class differences hidden in local spectral bands, by

$$\boldsymbol{\mathcal{F}}^{\text{diff}}_{\text{spec}} = \text{Conv1D}_{k_s}\left(\boldsymbol{\mathcal{F}}_{\text{spec}}\right) - \text{Conv1D}_{k_l}\left(\boldsymbol{\mathcal{F}}_{\text{spec}}\right). \quad (2)$$

Subsequently, the differential features are fed into a $1\times 1$ Conv1D to generate the spectral attention weights $\boldsymbol{\gamma}_s \in \mathbb{R}^{B\times C\times S}$ and the spectral bias parameters $\boldsymbol{\beta}_s \in \mathbb{R}^{B\times C\times S}$ as

$$\boldsymbol{\gamma}_s = 2\sigma\left(\boldsymbol{W}_\gamma \boldsymbol{\mathcal{F}}^{\text{diff}}_{\text{spec}}\right), \quad \boldsymbol{\beta}_s = \boldsymbol{W}_\beta \boldsymbol{\mathcal{F}}^{\text{diff}}_{\text{spec}} \quad (3)$$

where $\sigma(\cdot)$ is the sigmoid function, and $\boldsymbol{W}_\gamma$ and $\boldsymbol{W}_\beta$ denote learnable projections.

The spatial attention branch is designed to emphasize discriminative spatial regions. Symmetric to the spectral branch, we first obtain a spatial feature map $\boldsymbol{\mathcal{F}}_{\text{spa}} \in \mathbb{R}^{B\times C\times H\times W}$. Next, parallel Conv2D layers with kernel sizes $3\times 3$ and $7\times 7$ are employed, followed by differential enhancement $\boldsymbol{\mathcal{F}}^{\text{diff}}_{\text{spa}}$. Similarly, the spatial scaling $\boldsymbol{\gamma}_{hw} \in \mathbb{R}^{B\times C\times H\times W}$ and bias parameters $\boldsymbol{\beta}_{hw} \in \mathbb{R}^{B\times C\times H\times W}$ are obtained. Finally, SSAD fuses the calibration results in a residual form with scaling and shifting, yielding the calibrated feature map, defined as

$$\boldsymbol{\mathcal{X}}_{\text{out}} = \boldsymbol{\mathcal{X}} \odot \boldsymbol{\gamma}_s \odot \boldsymbol{\gamma}_{hw} + \boldsymbol{\beta}_s + \boldsymbol{\beta}_{hw} \in \mathbb{R}^{B\times C\times S\times H\times W} \quad (4)$$

where $\odot$ denotes element-wise multiplication.

b) *Dual-stream gated fusion module:* Given the complex geomorphology of coastal wetlands, vegetation patches often exhibit irregular shapes like slender structures and extended boundaries. To comprehensively represent these features, a global context modeling approach is proposed to complement local feature extraction.

As depicted in Fig. 4(b), the state space gating cell (SSGCell) is designed based on the selective state space model (SSM) framework. Specifically, we implement the cell architecture by leveraging the Mamba operator to establish an efficient modeling paradigm. By utilizing its $O(N)$ linear complexity and inherent selection mechanism [17], the SSGCell can effectively aggregate long-range spatial spectral dependencies while dynamically suppressing redundant background noise, which is more suitable for complex landscape classification in coastal wetlands. In general, SSM can be written as

$$\begin{aligned}\dot{\boldsymbol{h}}(t) &= \boldsymbol{A}(\boldsymbol{u}(t))\,\boldsymbol{h}(t) + \boldsymbol{B}(\boldsymbol{u}(t))\,\boldsymbol{u}(t)\\ \boldsymbol{y}(t) &= \boldsymbol{C}(\boldsymbol{u}(t))\,\boldsymbol{h}(t)\end{aligned} \quad (5)$$

Here, $\boldsymbol{h}(t)$ denotes the hidden state, and $\boldsymbol{u}(t)$ denotes the input sequence. The system matrices $\boldsymbol{A}(\cdot)$, $\boldsymbol{B}(\cdot)$, and $\boldsymbol{C}(\cdot)$ are modulated by the input. After discretization, the model can be written in the following recurrent form:

$$\begin{aligned}\boldsymbol{h}_{t+1} &= \bar{\boldsymbol{A}}(\boldsymbol{u}_t)\,\boldsymbol{h}_t + \bar{\boldsymbol{B}}(\boldsymbol{u}_t)\,\boldsymbol{u}_t\\ \boldsymbol{y}_t &= \bar{\boldsymbol{C}}(\boldsymbol{u}_t)\,\boldsymbol{h}_t\end{aligned} \quad (6)$$

where $\bar{\boldsymbol{A}}(\cdot)$, $\bar{\boldsymbol{B}}(\cdot)$ and $\bar{\boldsymbol{C}}(\cdot)$ denote the input-dependent discrete state transition, input-to-state, and state-to-output matrices, respectively. Building on this formulation, Mamba explicitly parameterizes $\bar{\boldsymbol{A}}$, $\bar{\boldsymbol{B}}$, and $\bar{\boldsymbol{C}}$ via convolutional and gating operations.

SSGCell first reshapes the input feature into a sequence representation $\boldsymbol{\mathcal{X}}' \in \mathbb{R}^{B\times R\times C}$, where $R = H\times W$. To encode 2-D spatial positions, learnable relative positional embeddings $\boldsymbol{\mathcal{P}}_h \in \mathbb{R}^{1\times C\times 1\times H}$ and $\boldsymbol{\mathcal{P}}_w \in \mathbb{R}^{1\times C\times W\times 1}$ are incorporated to the sequence. They are broadcast and added to form a 2-D positional tensor $\boldsymbol{\mathcal{P}}_{2D} = \boldsymbol{\mathcal{P}}_h + \boldsymbol{\mathcal{P}}_w \in \mathbb{R}^{1\times C\times W\times H}$. Then, $\boldsymbol{\mathcal{P}}_{2D}$ is reshaped and permuted into the sequence form $\boldsymbol{\mathcal{P}}_{seq} \in \mathbb{R}^{1\times R\times C}$. The position-enhanced sequence is obtained by

$$\boldsymbol{\mathcal{X}} = \boldsymbol{\mathcal{X}}' + \boldsymbol{\mathcal{P}}_{seq} \in \mathbb{R}^{B\times R\times C}. \quad (7)$$

The Mamba operator is then applied to $\tilde{\boldsymbol{\mathcal{X}}}$ to produce $\boldsymbol{\mathcal{Z}}_{\text{mamba}} = \text{Mamba}(\tilde{\boldsymbol{\mathcal{X}}})$. To adaptively regulate the contribution of different spatial positions, $\tilde{\boldsymbol{\mathcal{X}}}$ is permuted to $(B, C, R)$ and fed into a gating network composed of a 1-D convolution followed by a sigmoid function to generate gating coefficients $\boldsymbol{\mathcal{G}}$. The final SSGCell output is obtained by element-wise modulation as

$$\boldsymbol{\mathcal{O}}_{\text{Cell}} = \boldsymbol{\mathcal{Z}}_{\text{mamba}} \odot \boldsymbol{\mathcal{G}}. \quad (8)$$

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The multihead self-attention (MHSA) [51] branch operates in parallel with the SSGCell branch, as illustrated in Fig. 4(c). Query $\boldsymbol{Q}$, key $\boldsymbol{K}$, and value $\boldsymbol{V}$ are generated via $1 \times 1$ convolutions and rearranged into the multihead form, with a consistent tensor shape of $\mathbb{R}^{B\times h\times d_k\times R}$, where $B$ is the batch size, $h$ denotes the number of attention heads (we set $h = 8$ in all experiments), $d_k = C/h$ represents the feature dimension of each single head, and $R = H \times W$ stands for the total spatial sequence length. We also introduce 2-D relative positional encodings $\mathcal{P}_h$ and $\mathcal{P}_w$, they are summed and reshaped into the sequence form $\mathcal{P}_{seq} \in \mathbb{R}^{1\times h\times R\times d_k}$. The positional bias is then obtained by multiplying this positional embedding with the query features, yielding $\boldsymbol{E}_{pos} = \mathcal{P}_{seq}\boldsymbol{Q} \in \mathbb{R}^{B\times h\times R\times R}$. The content similarity is computed as $\boldsymbol{E}_{\text{cont}} = \boldsymbol{Q}\boldsymbol{K}^{\top}$ and the final attention logits are $\boldsymbol{E} = \boldsymbol{E}_{\text{cont}} + \boldsymbol{E}_{\text{pos}}$. The MHSA output is then obtained by

$$\mathcal{O}_{\text{MHSA}} = \text{Softmax}\,(\boldsymbol{E})\,\boldsymbol{V} \in \mathbb{R}^{B\times C\times H\times W}. \tag{9}$$

The gated fusion module is a key integration component, as illustrated in Fig. 4(d). The model first splits the feature map along the channel dimension, processing one half with SSGCell and the other with MHSA. To further exploit their complementary strengths, we introduce lightweight branch-aware attention enhancement, where channel-wise attention is applied to the SSGCell branch output to strengthen channel discriminability, and spatial-wise attention is applied to the MHSA branch output to emphasize spatial structure cues. Their outputs are then adaptively fused using a gating mechanism, yielding a robust integrated representation

$$\begin{aligned}\mathcal{Y} = {} & \mathcal{A}_c \odot (\mathcal{O}_{\text{Cell}} + \mathcal{C}\,(\mathcal{O}_{\text{Cell}})) \\ & + \mathcal{A}_m \odot (\mathcal{O}_{\text{MHSA}} + \mathcal{M}\,(\mathcal{O}_{\text{MHSA}}))\end{aligned} \tag{10}$$

where $\mathcal{Y} \in \mathbb{R}^{B\times C\times H\times W}$ and $\mathcal{A}_c, \mathcal{A}_m \in [0,1]^{B\times 1\times H\times W}$ denote spatially adaptive gate maps generated by a $1 \times 1$ convolution followed by a sigmoid activation, and $\mathcal{C}(\cdot)$ and $\mathcal{M}(\cdot)$ denote channel-wise and spatial-wise attention operators, respectively.

### B. Cross-Domain Source and Target FSL

The embedding features are fed into the FSL component to carry out classification tasks. We alternately execute FSL tasks for the SD and TD in accordance with the same procedural setup. Taking SD as an example, for class $k$, the samples in $\mathcal{S}$ are mapped by the feature extractor to an embedding set $\mathcal{Z}_k = \{z_i = f_\phi(\boldsymbol{x}_i) \mid (\boldsymbol{x}_i, y_i) \in \mathcal{S},\ y_i = k\}$, where $f_\phi$ denotes the feature extraction network. Conventional methods directly compute the mean of these embeddings as the initial prototype of the class by

$$\boldsymbol{p}_k^{(0)} = \frac{1}{|\mathcal{Z}_k|}\sum_{z\in\mathcal{Z}_k} z. \tag{11}$$

However, coastal wetlands are characterized by highly complex ecological transition zones, which inevitably lead to severe boundary ambiguity that severely skews traditional mean prototypes. Therefore, we propose DPIC, whose core

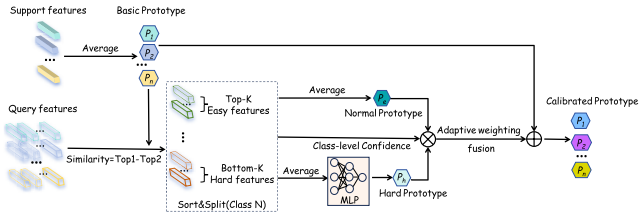



Fig. 5. Schematic of dynamic prototype calibration module.

**Algorithm 1** Few-Shot Learning

**Input:** $\mathcal{S} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{|\mathcal{S}|}$, $\mathcal{Q} = \{(\boldsymbol{x}_q, y_q)\}_{q=1}^{|\mathcal{Q}|}$;
**Output:** $\mathcal{L}^{\text{fsl}}$

$\mathcal{Z}_k = \{z_i = f_\phi(\boldsymbol{x}_i) \mid (\boldsymbol{x}_i, y_i) \in \mathcal{S},\ y_i = k\}$
$\boldsymbol{p}_k^{(0)} = \frac{1}{|\mathcal{Z}_k|}\sum_{z\in\mathcal{Z}_k} z, \quad \forall k \in \{1,\ldots,C\}$
$\mathcal{Z}_Q = \{z_q = f_\phi(\boldsymbol{x}_q) \mid (\boldsymbol{x}_q, y_q) \in \mathcal{Q}\}$
$\ell_k(z_q) = -\|z_q - \boldsymbol{p}_k^{(0)}\|_2^2$, $\hat{y}_q = \arg\max_k \ell_k(z_q)$
$m(z_q) = \ell_{(1)}(z_q) - \ell_{(2)}(z_q)$ for each $z_q$
$m_{\min} = \min_{z_q\in\mathcal{Z}_Q} m(z_q), \quad m_{\max} = \max_{z_q\in\mathcal{Z}_Q} m(z_q)$
**for** $k = 1,\ldots,C$ **do**
  $\mathcal{Q}_k = \{z_q \in \mathcal{Z}_Q \mid \hat{y}_q = k\}$
  $\mathcal{E}_k = \underset{\mathcal{U}\subseteq\mathcal{Q}_k,\ |\mathcal{U}|=K_{\text{typ}}}{\arg\max} \sum_{z\in\mathcal{U}} m(z)$
  $\mathcal{H}_k = \underset{\mathcal{U}\subseteq\mathcal{Q}_k,\ |\mathcal{U}|=K_{\text{hrd}}}{\arg\min} \sum_{z\in\mathcal{U}} m(z)$
  $\bar{\boldsymbol{p}}_k^N = \frac{1}{|\mathcal{E}_k|}\sum_{z\in\mathcal{E}_k} z$
  $\bar{\boldsymbol{p}}_k^H = \text{MLP}_{\text{hard}}\left(\frac{1}{|\mathcal{H}_k|}\sum_{z\in\mathcal{H}_k} z\right)$
  $m_k^{\text{avg}} = \frac{1}{|\mathcal{Q}_k|}\sum_{z\in\mathcal{Q}_k} m(z)$
  $\bar{m}_k = \frac{m_k^{\text{avg}} - m_{\min}}{m_{\max} - m_{\min}}$
  $s_{\text{sup}} = \bar{m}_k,\ s_{\text{nor}} = 1 - |\bar{m}_k - \mu|,\ s_{\text{hard}} = 1 - \bar{m}_k$
  $w_i = \frac{\exp(\tau\cdot s_i)}{\sum_j \exp(\tau\cdot s_j)}, \quad i,j \in \{\text{sup},\ \text{nor},\ \text{hard}\}$
  $\tilde{\boldsymbol{p}}_k = w_{\text{sup}}\boldsymbol{p}_k^{(0)} + w_{\text{nor}}\bar{\boldsymbol{p}}_k^N + w_{\text{hard}}\bar{\boldsymbol{p}}_k^H$
  $\alpha_k = \alpha_{\max}(1 - \bar{m}_k)$
  $\boldsymbol{p}_k^* = (1-\alpha_k)\boldsymbol{p}_k^{(0)} + \alpha_k\tilde{\boldsymbol{p}}_k$
**end for**
$\ell_k^*(z_q) = -\|z_q - \boldsymbol{p}_k^*\|_2^2$
$\mathcal{L}^{\text{fsl}} = \frac{1}{|\mathcal{Q}|}\sum_{(\boldsymbol{x}_q,y_q)\in\mathcal{Q}} \text{CE}(\{\ell_k^*(z_q)\}_{k=1}^C,\ y_q)$
**return** $\mathcal{L}^{\text{fsl}}$

procedure is illustrated in Fig. 5. The detailed procedure of the FSL is summarized in Algorithm 1. Specifically, we first use the initial prototypes $\{\boldsymbol{p}_k^{(0)}\}$ to conduct a metric-based prediction on the $\mathcal{Q}$ and compute the predictive confidence for each query sample. To evaluate the classification difficulty of each query embedding $z_q$, we focus on the top-2 logits of its prediction. We denote the highest logit as $\ell_1(z_q)$ and the second-highest logit as $\ell_2(z_q)$, and define the classification margin by

$$m\left(z_q\right) = \ell_{(1)}\left(z_q\right) - \ell_{(2)}\left(z_q\right). \tag{12}$$

A larger margin indicates that the sample lies farther from the decision boundary and its class assignment is more certain. For each class $k$, let the set of its query samples be $\mathcal{Q}_k = \{z_q \mid$

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$\hat{y}_q = k\}$. We then sort the samples in $\mathcal{Q}_k$ by $m(\boldsymbol{z}_q)$, select the high-margin typical samples to form the set $\mathcal{E}_k$, and select the low-margin hard samples to form the set $\mathcal{H}_k$. Accordingly, two auxiliary prototypes are constructed by

$$\bar{\boldsymbol{p}}_k^N = \frac{1}{|\mathcal{E}_k|}\sum_{z\in\mathcal{E}_k} z$$
$$\bar{\boldsymbol{p}}_k^H = \mathrm{MLP}_{\mathrm{hard}}\left(\frac{1}{|\mathcal{H}_k|}\sum_{z\in\mathcal{H}_k} z\right) \tag{13}$$

where $\bar{\boldsymbol{p}}_k^N$ represents the typical class center of class $k$. While $\bar{\boldsymbol{p}}_k^H$ is obtained by aggregating low-confidence hard samples.

For each class $k$, we construct a triplet of prototypes $\{\boldsymbol{p}_k^{(0)}, \bar{\boldsymbol{p}}_k^N, \bar{\boldsymbol{p}}_k^H\}$. To characterize class-level difficulty, we normalize the margins $\{m(\boldsymbol{z}_q)\}$ of all query samples in the current episode to [0, 1], and use the mean normalized margin of class $k$ as the class-level difficulty indicator $\bar{m}_k$; a larger $\bar{m}_k$ implies that the class is overall easier to discriminate. We define $s_{\mathrm{sup}} = \bar{m}_k$, $s_{\mathrm{nor}} = 1 - |\bar{m}_k - \mu|$, $s_{\mathrm{hard}} = 1 - \bar{m}_k$, where $\mu = 0.5$ is used to encourage the typical-sample prototype to play a larger role for classes with moderate difficulty. The fusion weights are computed via a softmax function with a scaling factor $\tau$ to recalibrate the importance of each component, by

$$w_i = \frac{\exp(\tau \cdot s_i)}{\sum_{j\in\{\mathrm{sup,nor,hard}\}} \exp(\tau \cdot s_j)} \tag{14}$$

where $\tau = 4$. The three prototypes are then fused to obtain a candidate calibrated prototype by

$$\tilde{\boldsymbol{p}}_k = w_{\mathrm{sup}}\boldsymbol{p}_k^{(0)} + w_{\mathrm{nor}}\bar{\boldsymbol{p}}_k^N + w_{\mathrm{hard}}\bar{\boldsymbol{p}}_k^H. \tag{15}$$

To mitigate the influence of noisy query predictions for easy classes, we further introduce a class-level adaptive interpolation strategy to produce the final calibrated prototype, defined as

$$\alpha_k = \alpha_{\max}(1 - \bar{m}_k)$$
$$\boldsymbol{p}_k^* = (1 - \alpha_k)\boldsymbol{p}_k^{(0)} + \alpha_k\tilde{\boldsymbol{p}}_k. \tag{16}$$

In this article, $\alpha_{\max}$ is the upper bound of interpolation (set to 0.3 in this article). This bounded interpolation prevents unreliable query predictions from dominating the prototype update, while still allowing hard and typical samples to provide class-structure information.

Finally, the calibrated prototypes $\boldsymbol{p}_k^*$ are used to recompute the query logits as follows:

$$\ell_k(\mathbf{z}_q) = -\|\mathbf{z}_q - \boldsymbol{p}_k^*\|_2^2. \tag{17}$$

Within each episode, FSL is performed by computing the Euclidean distances between query embeddings and the calibrated prototypes of all classes, and the model is optimized by minimizing the prediction loss on the query set. The FSL classification loss is defined by

$$\mathcal{L}^{fsl} = \frac{1}{|\mathcal{Q}|}\sum_{(x_q,y_q)\in\mathcal{Q}} \mathrm{CE}\left(\left\{\ell_k(\mathbf{z}_q)\right\}_{k=1}^C, y_q\right). \tag{18}$$

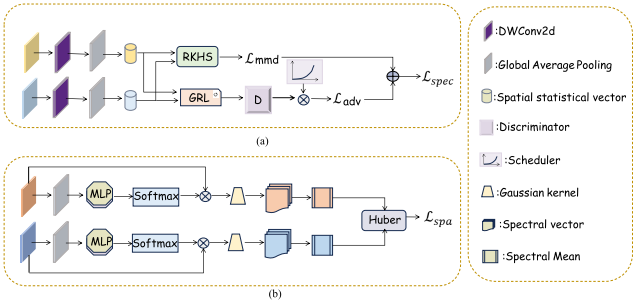



Fig. 6. Detailed architecture of the spatial and spectral alignment modules in the proposed SGDA framework. (a) Spatial alignment. (b) Spectral alignment.

**Algorithm 2** Synergistic Global Domain Alignment

**Input:** $\mathcal{F}^s_{\mathrm{map}}, \mathcal{F}^t_{\mathrm{map}} \in \mathbb{R}^{B\times C_{\mathrm{map}}\times H\times W}$, $\boldsymbol{v}_s, \boldsymbol{v}_t \in \mathbb{R}^{B\times C_{\mathrm{map}}}$
**Output:** $\mathcal{L}^{\mathrm{DA}}$

$\tilde{\mathcal{F}}^s_{\mathrm{map}} = \mathcal{F}^s_{\mathrm{map}} * \mathcal{W}^{(c)}_{dw}$, $\tilde{\mathcal{F}}^t_{\mathrm{map}} = \mathcal{F}^t_{\mathrm{map}} * \mathcal{W}^{(c)}_{dw}$
$\boldsymbol{a}_s = \frac{1}{H\times W}\sum \tilde{\mathcal{F}}^s_{\mathrm{map}}$, $\boldsymbol{a}_t = \frac{1}{H\times W}\sum \tilde{\mathcal{F}}^t_{\mathrm{map}}$
$\mathcal{L}_{\mathrm{mmd}} = \|\mathbb{E}[\varphi(\boldsymbol{a}_s)] - \mathbb{E}[\varphi(\boldsymbol{a}_t)]\|^2_{\mathcal{F}}$
$\mathcal{L}_{\mathrm{adv}} = -\mathbb{E}_{\boldsymbol{a}_s}[\log D_{\mathrm{spa}}(\boldsymbol{a}_s)] - \mathbb{E}_{\boldsymbol{a}_t}[\log(1 - D_{\mathrm{spa}}(\boldsymbol{a}_t))]$
$\lambda(t) = \frac{2}{1+\exp(-\gamma\cdot\frac{t}{T})} - 1$
Update $\mathcal{L}_{\mathrm{spa}}$ by $\lambda_{\mathrm{mmd}}\mathcal{L}_{\mathrm{mmd}} + \lambda(t)\mathcal{L}_{\mathrm{adv}}$
$\boldsymbol{w}_d = \mathrm{softmax}(\mathrm{MLP}(\mathrm{Norm}(\boldsymbol{v}_d)))$, $\tilde{\boldsymbol{v}}_d = \boldsymbol{w}_d \odot \boldsymbol{v}_d$
$\tilde{\boldsymbol{v}}_{s,\mathrm{low}} = \tilde{\boldsymbol{v}}_s * \boldsymbol{g}$, $\tilde{\boldsymbol{v}}_{t,\mathrm{low}} = \tilde{\boldsymbol{v}}_t * \boldsymbol{g}$
$\bar{\tilde{\boldsymbol{v}}}_{s,\mathrm{low}} = \mathbb{E}[\tilde{\boldsymbol{v}}_{s,\mathrm{low}}]$, $\bar{\tilde{\boldsymbol{v}}}_{t,\mathrm{low}} = \mathbb{E}[\tilde{\boldsymbol{v}}_{t,\mathrm{low}}]$
Update $\mathcal{L}_{\mathrm{spec}}$ by $\mathrm{Huber}(\bar{\tilde{\boldsymbol{v}}}_{s,\mathrm{low}}, \bar{\tilde{\boldsymbol{v}}}_{t,\mathrm{low}})$
$\mathcal{L}^{\mathrm{DA}} = \lambda_{\mathrm{spa}}\mathcal{L}_{\mathrm{spa}} + \lambda_{\mathrm{spec}}\mathcal{L}_{\mathrm{spec}}$
Insert $\mathcal{L}^{\mathrm{DA}}$ into the overall objective $\mathcal{L}_{\mathrm{total}}$ to update model parameters.
**return** $\mathcal{L}^{\mathrm{DA}}$

### C. SGDA Module

Since domain shift can severely degrade the classification performance, DA has been widely adopted in recent years [52]. Existing cross-domain alignment methods mostly rely on single-granularity constraints, which struggle to achieve sufficient domain invariance and often damage the class-discriminative feature structure in heterogeneous wetland scenarios. To overcome this limitation, we design an SGDA module that jointly calibrates spatial distribution discrepancies and spectral baseline shifts. The detailed architecture of SGDA is illustrated in Fig. 6. The process of SGDA is shown in Algorithm 2.

a) *Global spatial distribution alignment:* Considering that each channel corresponds to an independent spectral band with its own spatial response pattern, we take the band-mapped features as the shallow spatial feature maps $\mathcal{F}_{\mathrm{map}} \in \mathbb{R}^{B\times C_{\mathrm{map}}\times H\times W}$, where $C_{\mathrm{map}} = 100$ is the unified spectral dimension after projection. A depthwise convolution is applied to enhance channel-specific spatial details without disrupting spectral independence, as

$$\tilde{\mathcal{F}}^{(c)}_{\mathrm{map}} = \mathcal{F}^{(c)}_{\mathrm{map}} * \mathcal{W}^{(c)}_{\mathrm{dw}} \tag{19}$$

where $*$ denotes the 2-D convolution operation, and $\mathcal{W}^{(c)}_{dw} \in \mathbb{R}^{1\times 1\times k\times k}$ ($k = 3$) is the depthwise convolution kernel for the $c$th channel. The full kernel tensor $\mathcal{W}_{dw}$

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has a shape of $\mathbb{R}^{C_{\text{map}}\times 1\times k\times k}$. To cut off the forced semantic binding brought by local position coordinates, we employ global average pooling to compress the 2-D spatial topology into a 1-D global spatial statistical vector $\boldsymbol{a}$. Its $c$th element $\boldsymbol{a}_c \in \mathbb{R}^{B\times C_{\text{map}}}$ is computed by

$$a_c = \frac{1}{H\times W}\sum_{h=1}^{H}\sum_{w=1}^{W}\tilde{\mathcal{F}}_{\text{map}}(c,h,w). \tag{20}$$

To comprehensively eliminate spatial domain shifts caused by different sensors and environments, we introduce an explicit-implicit dual-engine collaborative alignment strategy. First, multikernel maximum mean discrepancy (MK-MMD) is introduced to map the source vector $\mathbf{a}_s$ and target vector $\mathbf{a}_t$ into a reproducing kernel Hilbert space (RKHS). By minimizing the marginal distribution discrepancy, it explicitly aligns the mean and variance of the two domains at a macroscopic scale

$$\mathcal{L}_{\text{mmd}} = \left\| \frac{1}{n_s}\sum_{i=1}^{n_s}\varphi\left(\boldsymbol{a}_s^{(i)}\right) - \frac{1}{n_t}\sum_{j=1}^{n_t}\varphi\left(\boldsymbol{a}_t^{(j)}\right)\right\|_{\mathcal{F}}^2 \tag{21}$$

where $\varphi(\cdot)$ denotes an implicit kernel mapping, and $\mathcal{F}$ is the corresponding RKHS.

We further introduce a spatial domain discriminator $D_{\text{spa}}(\cdot)$ equipped with a gradient reversal layer (GRL) for adversarial learning. The discriminator strives to distinguish the domain origin of the spatial vectors, while the feature extractor aims to confuse the origin, thereby implicitly calibrating highly complex nonlinear spatial style discrepancies

$$\mathcal{L}_{\text{adv}} = -\mathbb{E}_{\boldsymbol{a}_s}\left[\log D_{\text{spa}}(\boldsymbol{a}_s)\right] - \mathbb{E}_{\boldsymbol{a}_t}\left[\log\left(1 - D_{\text{spa}}(\boldsymbol{a}_t)\right)\right]. \tag{22}$$

$D(\cdot) \in (0,1)$ denotes the probability that a sample originates from SD. We further design an adaptive alignment-strength scheduler. This scheduler adjusts the weight of the adversarial loss according to training progress, emphasizing representation learning at the beginning and gradually increasing alignment strength as training proceeds, by

$$\lambda(t) = \frac{2}{1+\exp\left(-\gamma\cdot\frac{t}{T}\right)} - 1 \tag{23}$$

where $t$ is the current training iteration, $T$ is the total number of iterations, and $\gamma$ is a hyperparameter controlling the scheduling rate. Thus, the global spatial distribution alignment term is defined as

$$\mathcal{L}_{\text{spa}} = \lambda_{\text{mmd}}\mathcal{L}_{\text{mmd}} + \lambda(t)\,\mathcal{L}_{\text{adv}} \tag{24}$$

where $\lambda_{\text{mmd}}$ is a fixed hyperparameter balancing the MMD loss.

b) *Spectral characteristic alignment:* As shown in Fig. 6(b), to reduce the interference of spatial texture perturbations on spectral matching, we first apply global average pooling over the spatial dimensions to aggregate local spatial information into spectral vectors $\boldsymbol{v}_s$ and $\boldsymbol{v}_t$. We design a sample-adaptive dynamic band-weighting mechanism in which, after normalizing each sample's spectral vector, a lightweight network is used to generate band-importance weights, which are then applied to recalibrate the spectral responses as

$$\boldsymbol{w} = \text{softmax}\left(\text{MLP}\left(\text{Norm}\left(\boldsymbol{v}\right)\right)\right), \quad \tilde{\boldsymbol{v}} = \boldsymbol{w}\odot\boldsymbol{v}. \tag{25}$$

Considering that the SD and TD possess different category distributions, forcibly aligning the detailed morphological fluctuations of spectral curves would inevitably lead to severe negative transfer. Therefore, our alignment strategy strictly avoids disrupting the class-specific morphological characteristics. Instead, it focuses exclusively on the overall macroscopic brightness and complex background radiation shifts, which predominantly manifest as continuous, low-frequency baseline envelopes. To explicitly extract this global brightness baseline, we employ a 1-D Gaussian smoothing kernel $\boldsymbol{g}$ with a window size of $k = 7$ to filter the weighted spectral vector $\tilde{\boldsymbol{v}}_d$, obtaining the low-frequency envelope $\tilde{\boldsymbol{v}}_{d,\text{low}} \in \mathbb{R}^{C_{\text{map}}}$. By imposing consistency constraints solely on these macroscopic baselines, we calibrate the global spectral brightness shift while preserving the high-frequency morphological fluctuations critical for distinguishing distinct ground objects. The baseline alignment loss is defined by

$$\mathcal{L}_{\text{spec}} = \text{Huber}\left(\bar{\tilde{\boldsymbol{v}}}_{s,\text{low}},\ \bar{\tilde{\boldsymbol{v}}}_{t,\text{low}}\right) \tag{26}$$

where $\bar{\tilde{\boldsymbol{v}}}_{d,\text{low}}$ denotes the empirical mean vector of $\tilde{\boldsymbol{v}}_{d,\text{low}}$ across all samples.

Finally, the overall optimization objective is given by

$$\mathcal{L} = \mathcal{L}_{\text{FSL}} + \lambda_{\text{spec}}\mathcal{L}_{\text{spec}} + \lambda_{\text{spa}}\mathcal{L}_{\text{spa}} \tag{27}$$

where $\lambda_{\text{spec}}$ and $\lambda_{\text{spa}}$ denote the weighting coefficients for the two loss terms, respectively. The final optimization objective combines the episodic few-shot loss $\mathcal{L}^{fsl}$ with the DA loss $\mathcal{L}^{DA}$ terms using fixed weights selected by the parameter sensitivity analysis.

Based on this overall optimization objective, the detailed execution flow of the ACFSL pipeline is presented in Algorithm 3.

## IV. Experimental Results and Discussion

### A. Area of Study and Datasets

*Yellow River Delta:* The study area is located in the Yellow River Delta National Nature Reserve, Shandong Province, China (118°33'E–119°20'E and 37°35'N–38°12'N). The dominant vegetation types include *Robinia pseudoacacia (RP)*, tamarisks, natural willow (NW) forests, reeds, and so on. Fig. 7 shows three regions for monitoring wetland vegetation coverage using HSIC techniques [53].

*SD Dataset:* We used the Chikusei (CKS) dataset [54] as SD. The SD dataset was acquired on July 29, 2014, over agricultural and urban areas in Chikusei, Ibaraki, Japan, using a Headwall Hyperspec-VNIR-C imaging sensor. It contains 128 spectral bands spanning 363–1018 nm, with an image size of 2517 × 2335 pixels. The spatial resolution is 2.5 m, and the ground-truth annotations include 19 classes.

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**Algorithm 3** ACFSL

**Input:** Source dataset $\mathcal{D}_s$, Target labeled dataset $\mathcal{D}_t^{\text{train}}$, Target testing set $\mathcal{D}_t^{\text{test}} = \{(\boldsymbol{x}_c, c)\}$
**Output:** Target full-scene classification map $\boldsymbol{M} \in \mathbb{N}^{H \times W}$.

**for** $t = 1, \ldots, T$ **do**
  **if** $t$ is odd **then**
    $\mathcal{T}_s = \{\mathcal{S}_s, \mathcal{Q}_s\}$ from $\mathcal{D}_s$
    $\mathcal{L}_{\text{fsl}}^s$ = **Algorithm 1** $(\mathcal{T}_s; f_\phi, \Theta)$
    Update network: $\Theta \leftarrow \Theta - \eta \nabla_\Theta \mathcal{L}_{\text{fsl}}^s$
  **else**
    $\mathcal{T}_t = \{\mathcal{S}_t, \mathcal{Q}_t\}$ from $\tilde{\mathcal{D}}_t^{\text{train}}$
    $\mathcal{L}_{\text{fsl}}^t$ = **Algorithm 1** $(\mathcal{T}_t; f_\phi, \Theta)$
    Update network: $\Theta \leftarrow \Theta - \eta \nabla_\Theta \mathcal{L}_{\text{fsl}}^t$
  **end if**
  $\mathcal{L}^{\text{DA}}$ = **Algorithm 2** $(\mathcal{T}_s, \mathcal{T}_t; f_\phi, \Theta)$
  Update network: $\Theta \leftarrow \Theta - \eta \nabla_\Theta \mathcal{L}^{\text{DA}}$
**end for**
$\boldsymbol{P}^*$ = **Algorithm 1** $(\tilde{\mathcal{D}}_t^{\text{train}}; f_\phi, \Theta)$
**for** each instance $(\boldsymbol{x}_c, c) \in \mathcal{D}_t^{\text{test}}$ **do**
  $\boldsymbol{M}(c) = \arg\min_{k \in C} \|f_\phi(\boldsymbol{x}_c) - \boldsymbol{p}_k^*\|_2^2$
**end for**
**return** $\boldsymbol{M}$

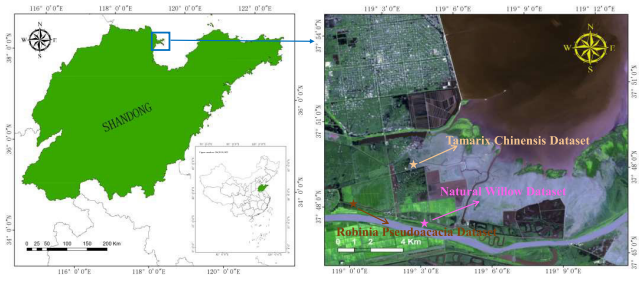



Fig. 7. Overview of the study area.

*TD Datasets:* We used the three self-constructed hyperspectral wetland vegetation datasets from the Yellow River Delta and one public dataset as TD.

1) *NW Dataset:* Focusing on NW forests, the NW dataset was captured in September 2022 using a DJI M600 Pro UAV equipped with a Cubert Utilities Firefly S185 camera. Radiometric calibration and geometric correction were performed using CORE and TORCH software. The data size is $590 \times 512 \times 126$, covering 0.4–0.95 $\mu$m, with a spatial resolution of 1 m.
2) *TC Dataset:* Processed in the same manner as the NW dataset, the *Tamarix Chinensis (TC)* dataset was captured on September 25, 2022. This dataset focuses on tamarisk forests and includes nine land cover types. The data size is $784 \times 1024 \times 126$, with a spatial resolution of 0.5 m.
3) *RP Dataset:* This dataset was acquired at 11:15 A.M. on June 21, 2023, using a DJI UAV, targeting RP forests. It contains 11 land cover types, with an image size of $1484 \times 1246$ pixels and 126 spectral bands, and a spatial resolution of 1 m.
4) *PU Dataset:* Acquired by ROSIS over Pavia University (PU), this dataset has an image size of $610 \times 340$ pixels with 1.3-m spatial resolution and 103 spectral channels. It covers nine land cover classes and contains 42 776 labeled samples.

Table I summarizes the ground-truth maps, the number of classes, land-cover types, and the number of labeled samples for each class in the five datasets.

### B. Experimental Setting

ACFSL is implemented in Python using the PyTorch framework, and all experiments are conducted on the same hardware platform equipped with an NVIDIA RTX 3090 GPU, 96 GB RAM, and an Intel Core i9-10900K CPU.

We denote by $C$ the number of classes in TD and by $K$ the number of samples per class in $\mathcal{S}$ of each episode. For the NW and TC datasets, $C = 9$; for the RP dataset, $C = 11$. Each meta-task is constructed as a $C$-way $K$-shot episode. During meta-training, $K$ is set to 1 for both $\mathcal{T}_s$ and $\mathcal{T}_t$. The $\mathcal{Q}$ is used to evaluate the learned metric space and classifier. For fair comparability, the number of $\mathcal{Q}$ samples per class is fixed to 19, which is consistent with DCFSL [27]. ACFSL follows an episode-based learning paradigm, alternately constructing $\mathcal{T}_s$ and $\mathcal{T}_t$ during training.

Furthermore, a disjoint sampling strategy is employed. Specifically, during the data sampling process in both the SD and TD, for every randomly selected training pixel $(x, y)$, we establish a $(2w + 1) \times (2w + 1)$ exclusion buffer zone around its spatial coordinate, where $2w + 1 = 9$ corresponds to our patch size. Any candidate pixels falling within this buffer zone are strictly masked out and prohibited from being included in the testing sets. The feature embedding dimension is set to 160. The network is optimized using Adam with a learning rate of $7e^{-4}$, and the total number of training iterations is set to 10 000. The weight $\lambda$ of the domain-adversarial loss is scheduled to increase progressively with training progress. Classification performance is evaluated using commonly adopted HSIC metrics, including overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (Kappa). In addition, we report the number of parameters, floating-point operations per second (FLOP), and training and testing times to comprehensively assess ACFSL. All experiments are independently repeated five times, and the results are reported in the form of "mean $\pm$ standard deviation."

### C. Comparison With Other Methods

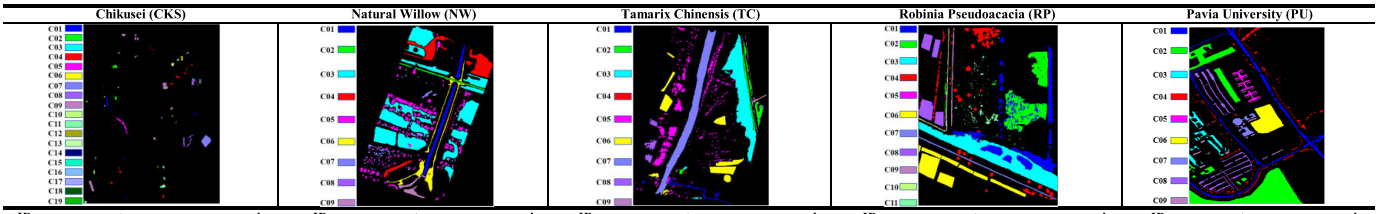
We conduct extensive experiments to compare ACFSL with nine state-of-the-art approaches. Specifically, WFCG [13], HyperSINet [55], and the baseline HyperSIGMA [56] are three end-to-end supervised deep learning methods. CMFSL [57] and CTF-SSCL [58] belong to FSL methods. For CDFSL, we select four representative methods, including DCFSL [27], Gia-CFSL [37], ADAFSL [36], and FDFSL [59]. All hyperparameter settings follow the original papers for a fair comparison.

Tables II–V report the accuracy metrics of the above methods on the four TD datasets. For all classification methods, the best results are highlighted in bold, and the second-best results are marked with an underline.

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TABLE I

GROUND TRUTH AND THE DETAILED NUMBERS OF THE SAMPLES FOR EACH CLASS IN BOTH SD AND TD DATASETS




| ID | types | samples | ID | types | samples | ID | types | samples | ID | types | samples | ID | types | samples |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Water | 2845 | 1 | Artificial channel | 4003 | 1 | Small tidal creek | 4435 | 1 | Phragmites australis | 106698 | 1 | Asphalt | 6631 |
| 2 | Bare soil(school) | 2859 | 2 | Highway | 1114 | 2 | Phragmites australis | 9860 | 2 | Robinia pseudoacacia | 93775 | 2 | Meadows | 18649 |
| 3 | Bare soil(park) | 286 | 3 | Phragmites australis | 32896 | 3 | Desiccated Spartime alterniflora | 52484 | 3 | Water | 133237 | 3 | Gravel | 2099 |
| 4 | Bare soil(farmland) | 48525 | 4 | Water | 9084 | 4 | Abandoned chinensis | 200 | 4 | Salix matsudana | 47836 | 4 | Trees | 3064 |
| 5 | Natural plants | 4297 | 5 | Willow | 8808 | 5 | Tamarix chinensis | 33744 | 5 | Tamarix chinensis | 869 | 5 | Painted metal sheets | 1345 |
| 6 | Weeds in farmland | 1108 | 6 | Earth road | 4982 | 6 | Suaeda salsa | 24505 | 6 | Rice paddy | 67437 | 6 | Bare Soil | 5029 |
| 7 | Forest | 20516 | 7 | Tamarix | 2909 | 7 | Large tidal creek | 47860 | 7 | Highway | 12583 | 7 | Bitumen | 1330 |
| 8 | Grass | 6515 | 8 | Cement | 118 | 8 | TC and SS mixed | 5648 | 8 | Abandoned farmland | 42830 | 8 | Self-Blocking Bricks | 3682 |
| 9 | Rice field(grown) | 13369 | 9 | Tidal flat | 1444 | 9 | Roads | 2891 | 9 | Firebreak ditch | 7878 | 9 | Shadows | 947 |
| 10 | Rice field(first stage) | 1268 | | Total | 65358 | | Total | 181627 | 10 | Dirt road | 7602 | | Total | 42776 |
| 11 | Row crops | 5961 | | | | | | | 11 | Withered Robinia pseudoacacia | 9236 | | | |
| 12 | Plastic house | 2193 | | | | | | | | Total | 529981 | | | |
| 13 | Manmade(non-dark) | 1220 | | | | | | | | | | | | |
| 14 | Manmade(dark) | 7664 | | | | | | | | | | | | |
| 15 | Manmade(blue) | 431 | | | | | | | | | | | | |
| 16 | Manmade(red) | 222 | | | | | | | | | | | | |
| 17 | ManMade grass | 1040 | | | | | | | | | | | | |
| 18 | Asphalt | 801 | | | | | | | | | | | | |
| 19 | Paved ground | 145 | | | | | | | | | | | | |
| | Total | 121265 | | | | | | | | | | | | |

TABLE II

COMPARISON OF CLASSIFICATION RESULTS FOR TEN METHODS ON THE NW DATASET (FIVE LABELED SAMPLES PER CLASS)

| Class | WFCG[13] | HyperSINet[56] | HyperSIGMA[57] | CMFSL[58] | CTF-SSCL[59] | Gia-CFSL[38] | DCFSL[27] | ADAFSL[36] | FDFSL[60] | ACFSL |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 89.13±1.26 | 91.84±5.45 | 94.65±1.78 | 92.76±2.71 | 95.97±0.82 | 94.28±2.54 | 95.82±0.71 | 94.79±1.81 | 93.62±5.03 | **96.32±0.61** |
| 2 | 87.25±10.93 | 86.76±1.80 | 75.14±9.30 | **92.57±4.45** | 87.62±9.54 | 87.68±7.89 | 83.45±13.02 | 85.31±10.48 | 88.29±8.62 | 84.62±9.31 |
| 3 | 81.78±5.72 | 71.02±23.12 | 50.01±11.79 | 82.14±14.42 | 79.88±9.39 | 80.67±5.02 | 79.79±14.13 | 77.68±7.91 | 77.54±8.56 | **82.73±7.18** |
| 4 | 75.92±9.35 | 90.16±4.78 | 86.78±5.46 | 91.92±3.00 | 93.89±1.82 | 93.98±1.83 | 94.01±5.34 | 91.88±2.91 | **94.36±1.48** | 94.27±1.98 |
| 5 | 62.19±5.18 | 62.39±9.06 | 71.69±8.11 | 57.26±10.45 | 74.79±10.27 | 60.12±8.11 | 68.24±13.92 | 66.78±12.12 | 63.05±11.03 | **76.89±10.09** |
| 6 | 90.30±2.64 | 85.36±6.09 | 75.30±7.15 | 87.92±3.98 | 87.42±6.41 | 85.41±5.89 | 82.87±9.12 | 86.93±9.32 | **91.13±1.89** | 86.91±7.23 |
| 7 | 39.04±3.74 | 18.55±7.86 | 44.58±12.31 | 64.27±11.95 | 69.54±5.63 | **72.29±4.77** | 54.31±16.54 | 49.82±17.23 | 72.12±4.79 | 67.87±4.58 |
| 8 | 63.27±2.90 | 51.73±15.02 | 85.95±4.89 | 85.37±7.34 | 88.89±10.55 | 88.97±6.52 | 81.47±10.89 | 74.56±10.98 | 88.64±6.65 | **90.53±1.91** |
| 9 | 89.79±1.33 | 71.06±18.26 | 88.37±6.25 | **92.65±2.97** | 90.72±6.92 | 89.52±7.71 | 91.09±5.54 | 91.54±3.29 | 89.98±5.04 | 90.46±5.42 |
| OA(%) | 77.19±6.15 | 65.15±13.28 | 60.05±5.74 | 80.58±1.07 | 80.31±0.09 | 79.96±3.85 | 80.12±7.48 | 78.79±1.49 | 79.75±3.38 | **83.05±4.12** |
| AA(%) | 75.41±2.71 | 69.87±7.32 | 74.72±5.49 | 82.98±1.29 | 84.73±2.75 | 83.66±1.38 | 81.23±4.15 | 78.47±5.23 | 84.30±2.95 | **85.62±1.28** |
| Kappa (*100) | 68.38±7.27 | 64.65±14.46 | 59.66±6.14 | 74.43±7.75 | 75.12±4.98 | 74.56±3.58 | 73.98±6.62 | 71.89±2.73 | 73.35±5.19 | **77.83±4.17** |

TABLE III

COMPARISON OF CLASSIFICATION RESULTS FOR TEN METHODS ON THE TC DATASET (FIVE LABELED SAMPLES PER CLASS)

| Class | WFCG[13] | HyperSINet[56] | HyperSIGMA[57] | CMFSL[58] | CTF-SSCL[59] | Gia-CFSL[38] | DCFSL[27] | ADAFSL[36] | FDFSL[60] | ACFSL |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 61.22±7.81 | 39.57±7.22 | 40.16±14.27 | 80.23±4.53 | 73.41±14.10 | **81.85±1.22** | 74.55±5.28 | 77.83±6.46 | 78.16±8.25 | 80.28±8.55 |
| 2 | 63.19±16.23 | 69.29±8.10 | 68.59±6.54 | 64.44±11.82 | 64.25±4.26 | 74.13±6.24 | 81.50±5.14 | 79.21±8.90 | 78.48±6.87 | **85.81±3.24** |
| 3 | 90.43±4.19 | 91.16±4.27 | 66.21±38.13 | 92.54±1.19 | 93.27±3.81 | 94.17±1.17 | **94.83±1.26** | 90.25±1.82 | 93.13±1.52 | 93.65±0.85 |
| 4 | 56.25±28.54 | 35.11±14.23 | 49.50±16.85 | 79.21±11.50 | 92.51±4.13 | 85.13±10.21 | 77.81±11.83 | 63.42±3.18 | 91.26±2.41 | **94.57±1.65** |
| 5 | 34.58±6.27 | 50.11±16.21 | 19.26±7.19 | 46.45±12.12 | 56.10±13.55 | **57.26±11.83** | 42.26±14.54 | 38.88±15.61 | 54.25±7.26 | 55.93±4.53 |
| 6 | 59.27±4.55 | 45.85±15.84 | 20.43±21.50 | 54.29±1.87 | 55.16±9.28 | **62.49±8.25** | 52.19±12.19 | 47.99±11.57 | 52.50±12.19 | 58.89±6.80 |
| 7 | 91.19±2.10 | 90.59±4.82 | 89.23±2.51 | 91.44±1.80 | 94.13±1.66 | **95.93±1.20** | 94.21±1.67 | 93.19±2.13 | 92.49±0.80 | 94.81±0.79 |
| 8 | 33.44±7.89 | 28.14±6.23 | 50.19±11.21 | **69.23±12.58** | 62.19±13.27 | 63.80±9.24 | 63.48±4.51 | 61.12±7.84 | 57.87±3.83 | 59.50±8.89 |
| 9 | 71.26±17.13 | 81.83±6.10 | 82.24±2.85 | 86.17±4.29 | 79.53±6.53 | **88.47±2.51** | 86.99±2.96 | 84.85±4.25 | 84.53±3.14 | 86.86±2.10 |
| OA(%) | 67.45±3.85 | 66.12±5.10 | 54.80±11.25 | 75.25±2.55 | 75.10±2.80 | 78.30±2.95 | 73.20±3.60 | 71.40±4.10 | 77.15±2.10 | **81.10±2.65** |
| AA(%) | 62.31±8.12 | 59.29±4.20 | 53.98±6.80 | 73.78±2.40 | 74.51±2.15 | 77.14±2.45 | 74.20±2.85 | 70.75±3.20 | 75.85±1.35 | **78.92±1.85** |
| Kappa (*100) | 61.10±4.25 | 59.60±5.50 | 45.15±11.80 | 67.40±3.15 | 69.50±3.20 | 72.10±3.30 | 68.40±4.10 | 66.20±4.80 | 74.05±2.30 | **76.25±2.80** |

Table II reports the results of all methods on the NW dataset. Evidently, the proposed ACFSL exhibits a clear performance advantage over the competing approaches, achieving an OA that is 2.47% higher than the second-best method. Notably, ACFSL delivers markedly better performance on several challenging classes. For example, due to class-imbalanced sample distributions, conventional supervised learning methods perform poorly on classes such as cement and tamarix, whereas ACFSL still attains 90.53% OA for cement. Furthermore, ACFSL achieves the best accuracy on four classes, indicating that its feature extractor can map the raw data into a more discriminative embedding space, thereby enabling clearer separation among different classes.

Table III lists the classification performance of the proposed method and the competing approaches on the TC dataset. It can be observed that ACFSL outperforms all baselines, surpassing the second-ranked method by 2.80%, 1.78%, and 2.20% in terms of OA, AA, and Kappa, respectively. Despite the evident class imbalance (e.g., only 200 samples are available for the abandoned chinensis class), ACFSL still achieves 94.57% OA. Table III also indicates that severe interclass confusion can cause noticeable degradation for some

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TABLE IV

COMPARISON OF CLASSIFICATION RESULTS FOR TEN METHODS ON THE RP DATASET (FIVE LABELED SAMPLES PER CLASS)

| Class | WFCG[13] | HyperSINet[56] | HyperSIGMA[57] | CMFSL[58] | CTF-SSCL[59] | Gia-CFSL[38] | DCFSL[27] | ADAFSL[36] | FDFSL[60] | ACFSL |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4.21±4.62 | 48.12±5.13 | 36.02±8.35 | 51.34±14.82 | 42.15±10.24 | 50.17±7.04 | 49.93±8.56 | 40.22±14.13 | 41.08±17.21 | **52.82±8.05** |
| 2 | 73.11±1.84 | 50.27±18.73 | 60.14±16.89 | 59.28±9.31 | 73.42±8.96 | 71.05±5.42 | 73.71±7.98 | 61.84±11.75 | 71.66±8.02 | **83.25±6.23** |
| 3 | 78.63±1.55 | 91.36±1.47 | 87.41±8.64 | 92.03±3.35 | 92.11±2.68 | 91.64±4.41 | **92.82±3.27** | 89.73±5.76 | 91.25±2.79 | 92.04±3.92 |
| 4 | **55.24±2.68** | 36.83±6.85 | 37.26±11.52 | 51.92±6.48 | 53.16±10.13 | 50.83±8.54 | 50.61±8.43 | 51.47±7.69 | 50.23±5.88 | 51.81±6.54 |
| 5 | 4.42±2.25 | 4.17±4.16 | 75.38±13.86 | 76.45±7.94 | 71.28±8.05 | 80.04±9.12 | 79.52±6.24 | 64.21±9.78 | 77.05±12.76 | **83.46±6.51** |
| 6 | **92.45±1.33** | 79.42±2.15 | 92.26±1.94 | 75.13±10.56 | 73.15±9.32 | 78.23±6.57 | 83.76±7.25 | 80.42±8.19 | 79.54±10.87 | 85.83±5.26 |
| 7 | 41.03±13.27 | 87.15±2.14 | 91.24±9.47 | 94.72±1.38 | 94.31±1.75 | 94.96±2.35 | 96.05±0.72 | 92.84±3.46 | 94.07±2.38 | **97.74±0.65** |
| 8 | 85.26±3.64 | 82.35±13.41 | 86.42±2.68 | 88.71±1.92 | 86.43±4.81 | 86.72±4.53 | 85.41±4.38 | 80.52±6.74 | 86.74±2.69 | **89.93±5.57** |
| 9 | 30.14±28.35 | 35.26±13.87 | 66.17±9.85 | 77.32±9.04 | 70.04±14.76 | 74.02±13.85 | 71.43±9.82 | **78.03±10.64** | 71.16±14.03 | 73.48±10.45 |
| 10 | 21.43±15.36 | 62.07±13.14 | 48.05±15.06 | 84.17±7.25 | 81.53±6.54 | 76.84±17.02 | 82.75±6.98 | 76.04±15.67 | **85.07±3.14** | 84.22±5.53 |
| 11 | 53.28±8.94 | 16.32±4.57 | 61.24±36.85 | 61.74±9.03 | 65.37±10.35 | 64.25±10.96 | 67.72±13.34 | 65.63±12.08 | 68.14±10.95 | **71.41±7.76** |
| OA(%) | 66.54±0.48 | 63.97±4.42 | 67.81±2.33 | 72.91±0.67 | 72.74±2.28 | 74.21±1.04 | 75.36±1.32 | 70.14±2.41 | 72.73±1.77 | **78.27±2.74** |
| AA(%) | 49.02±1.84 | 53.94±3.57 | 67.41±3.76 | 73.89±1.43 | 73.00±2.06 | 74.43±3.34 | 75.79±1.78 | 70.99±1.82 | 74.18±2.47 | **78.72±3.48** |
| Kappa (*100) | 59.82±0.41 | 57.74±4.88 | 62.43±2.31 | 67.26±3.78 | 68.05±2.41 | 69.43±1.26 | 70.91±1.47 | 63.82±2.45 | 67.48±1.92 | **73.14±2.96** |

TABLE V

COMPARISON OF CLASSIFICATION RESULTS FOR TEN METHODS ON THE PU DATASET (FIVE LABELED SAMPLES PER CLASS)

| Class | WFCG[13] | HyperSINet[56] | HyperSIGMA[57] | CMFSL[58] | CTF-SSCL[59] | Gia-CFSL[38] | DCFSL[27] | ADAFSL[36] | FDFSL[60] | ACFSL |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 5.45±9.89 | 64.86±11.86 | 32.28±20.37 | 69.98±4.19 | 42.67±18.74 | 76.05±5.78 | 70.89±4.45 | **81.73±1.20** | 77.36±6.10 | 80.02±1.91 |
| 2 | 57.32±12.04 | 68.03±6.37 | 54.18±8.13 | 71.98±12.26 | 54.18±13.12 | 79.12±2.36 | 74.93±11.81 | 70.15±13.44 | 76.56±9.04 | **88.37±3.41** |
| 3 | **81.86±6.11** | 75.91±12.33 | 78.16±7.97 | 59.89±10.37 | 35.12±17.24 | 53.68±8.83 | 60.03±11.49 | 58.97±8.36 | 57.06±10.82 | 54.88±10.01 |
| 4 | 49.86±31.47 | 73.78±8.86 | 59.28±3.88 | 87.32±3.38 | 81.76±7.13 | 86.16±3.43 | 91.57±2.20 | 83.45±5.16 | 84.49±1.55 | **92.61±1.57** |
| 5 | 78.27±22.21 | 44.87±19.84 | 94.81±4.23 | 95.37±0.12 | 95.97±1.44 | 89.75±2.78 | 93.98±2.33 | 93.82±1.42 | 89.99±1.22 | **97.26±0.47** |
| 6 | 30.58±35.82 | 43.47±22.03 | 75.22±17.92 | **75.54±13.02** | 40.56±23.52 | 69.89±10.12 | 68.67±12.15 | 69.09±11.63 | 60.18±13.26 | 67.23±9.62 |
| 7 | 71.69±17.08 | 68.02±12.98 | **89.56±5.83** | 77.97±4.72 | 60.29±30.91 | 68.98±2.31 | 79.08±3.26 | 75.56±3.43 | 78.82±3.90 | 62.18±4.04 |
| 8 | 37.91±9.15 | 44.35±23.66 | 34.23±12.74 | 67.79±10.93 | 48.29±11.16 | 65.89±10.47 | 59.04±13.48 | 67.54±13.51 | **71.62±12.31** | 69.54±11.37 |
| 9 | 30.94±13.54 | 70.06±6.24 | 36.38±16.14 | 91.97±0.42 | 88.21±9.40 | 93.57±2.32 | 94.94±0.95 | 93.56±0.42 | 91.37±1.97 | **95.02±0.13** |
| OA(%) | 44.95±5.21 | 65.48±4.23 | 52.03±3.17 | 75.72±2.35 | 53.21±1.51 | 75.76±2.45 | 75.82±5.63 | 75.87±5.58 | 76.38±3.39 | **81.70±2.88** |
| AA(%) | 49.32±2.55 | 61.48±1.98 | 61.57±1.64 | 77.53±3.84 | 60.78±3.58 | 75.90±3.01 | 77.01±1.25 | 77.10±1.79 | 76.49±2.56 | **78.56±2.56** |
| Kappa (*100) | 37.84±4.09 | 58.63±2.61 | 44.36±3.38 | 67.87±4.12 | 42.89±2.31 | 74.15±1.08 | 69.87±6.12 | 70.28±6.38 | 72.73±1.78 | **76.61±2.12** |

algorithms on several land-cover types; nevertheless, ACFSL improves OA by 7.90% over DCFSL and by 13.65% over WFCG. Moreover, the AA standard deviation of 1.85% suggests reduced sensitivity to data fluctuations, yielding more robust and stable classification results.

For the RP dataset, *Salix matsudana* and TC are highly intermingled spatially, making it the most challenging among the three datasets. Nevertheless, as shown in Table IV, ACFSL achieves the best performance. For the RP class, ACFSL attains an accuracy of 83.25%, which is 9.54% higher than the second-best result, mainly benefiting from the proposed DPIC and synergistic global alignment mechanisms. ACFSL achieves an accuracy of 83.46% on TC, further verifying its strong discriminative capability under severe interclass overlap and class imbalance.

On the public PU dataset, as reported in Table V, ACFSL continues to demonstrate significant superiority, achieving the highest OA of 81.70% and surpassing the second-best method by a notable margin of 5.32%. Notably, for the Shadows class, which typically suffers from severely limited samples in urban scenes, ACFSL still attains an accuracy of 95.02%. Furthermore, ACFSL achieves the best accuracy on four out of nine classes, indicating its superior capability in extracting robust representations and preserving class-discriminative structures. The relatively low standard deviation in OA (2.88%) further indicates its high stability.

Figs. 8–11 present the pseudocolor images and the corresponding classification maps for the TD datasets. From the visual results, the proposed ACFSL produces classification maps that are more consistent with the ground-truth annotations. Specifically, Fig. 8 shows that on the NW dataset, ACFSL yields overall smoother maps with noticeably reduced noise and exhibits high spatial consistency with the ground truth. Fig. 9 indicates that even on the TC dataset, where some

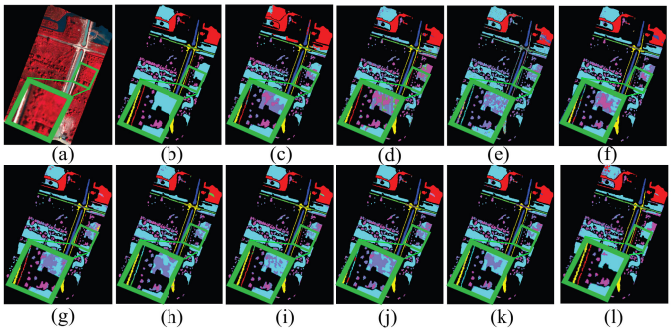



Fig. 8. Data visualization and the classification maps for the NW data obtained with different methods, including (a) false-color image of NW, (b) ground truth, (c) WFCG, (d) HyperSINet, (e) HyperSIGMA, (f) CMFSL, (g) CTF-SSCL, (h) Gia-CFSL, (i) DCFSL, (j) ADAFSL, (k) FDFSL, and (l) ACFSL.

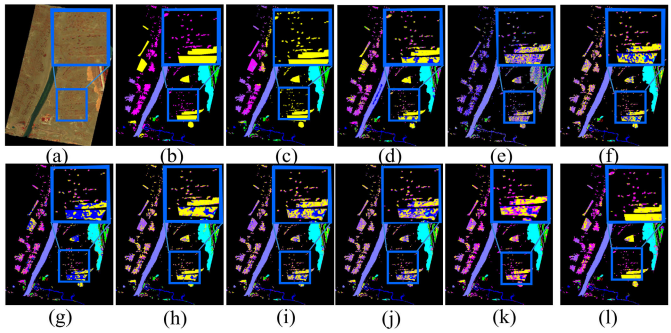



Fig. 9. Data visualization and the classification maps for the TC data obtained with different methods, including (a) false-color image of TC, (b) ground truth, (c) WFCG, (d) HyperSINet, (e) HyperSIGMA, (f) CMFSL, (g) CTF-SSCL, (h) Gia-CFSL, (i) DCFSL, (j) ADAFSL, (k) FDFSL, and (l) ACFSL.

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TABLE VI
COMPARISON WITH ALL METHODS ON OTHER METRICS

| | Methods | WFCG | HyperSINet | HyperSIGMA | CMFSL | CTF-SSCL | Gia-CFSL | DCFSL | ADAFSL | FDFSL | ACFSL |
|---|---|---|---|---|---|---|---|---|---|---|---|
| NW | Training time(s) | 145.98 | 60.44 | 75.95 | 583.21 | 431.39 | 2212.57 | 1925.30 | 1745.8 | 1186.30 | 1688.24 |
| | Testing time(s) | 2.35 | 0.01 | 159.32 | 3.89 | 2.67 | 3.97 | 7.82 | 4.57 | 10.38 | 7.51 |
| | Parameters(M) | 0.0620 | 0.1214 | 175.9761 | 0.6452 | 0.0797 | 0.2052 | 0.0372 | 0.0259 | 0.0565 | 0.0848 |
| | FLOPS(G) | 18.9441 | 13.7754 | 444.1928 | 0.2583 | 0.0519 | 5.6802 | 0.0621 | 0.0457 | 0.2049 | 0.3909 |
| TC | Training time(s) | 388.62 | 185.18 | 122.55 | 651.32 | 487.88 | 2354.42 | 2127.13 | 1874.96 | 2037.67 | 1796.32 |
| | Testing time(s) | 8.64 | 0.01 | 440.43 | 10.91 | 7.62 | 10.73 | 45.37 | 12.72 | 53.78 | 12.68 |
| | Parameters(M) | 0.0620 | 0.1214 | 175.9761 | 0.6449 | 0.0797 | 0.2051 | 0.0372 | 0.0259 | 0.0565 | 0.0848 |
| | FLOPS(G) | 59.20 | 43.0444 | 444.1928 | 0.2582 | 0.0519 | 5.68 | 0.0621 | 0.0457 | 0.2058 | 0.3909 |
| RP | Training time(s) | 236.63 | 107.74 | 106.14 | 728.85 | 508.62 | 2534.64 | 1030.98 | 2656.7 | 1742.7 | 3196.47 |
| | Testing time(s) | 4.27 | 0.01 | 323.04 | 7.89 | 5.39 | 8.03 | 7.93 | 9.26 | 30.95 | 14.15 |
| | Parameters(M) | 0.0622 | 0.1215 | 175.9761 | 0.6443 | 0.0798 | 0.1933 | 0.0366 | 0.0252 | 0.0559 | 0.0849 |
| | FLOPS(G) | 29.0489 | 21.1379 | 444.1928 | 0.2585 | 0.0063 | 5.65 | 0.0621 | 0.0459 | 0.2056 | 0.4778 |
| PU | Training time(s) | 104.40 | 40.68 | 90.52 | 373.34 | 389.91 | 1394.73 | 769.7 | 1345.08 | 1082.19 | 1929.06 |
| | Testing time(s) | 1.68 | 0.0035 | 104.12 | 5.59 | 1.62 | 2.33 | 2.30 | 2.74 | 7.21 | 3.18 |
| | Parameters(M) | 0.0590 | 0.1198 | 175.9761 | 0.6452 | 0.0797 | 0.2052 | 0.0372 | 0.0259 | 0.0565 | 0.0848 |
| | FLOPS(G) | 12.3769 | 9.1336 | 444.1928 | 0.2583 | 0.0052 | 5.6802 | 0.0621 | 0.0457 | 0.2049 | 0.3909 |

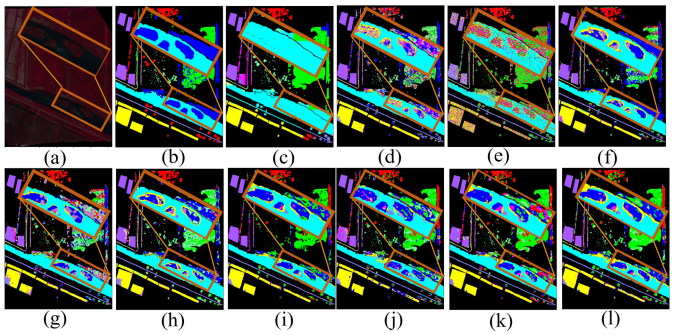



Fig. 10. Data visualization and the classification maps for the RP data obtained with different methods, including (a) false-color image of RP, (b) ground truth, (c) WFCG, (d) HyperSINet, (e) HyperSIGMA, (f) CMFSL, (g) CTF-SSCL, (h) Gia-CFSL, (i) DCFSL, (j) ADAFSL, (k) FDFSL, and (l) ACFSL.

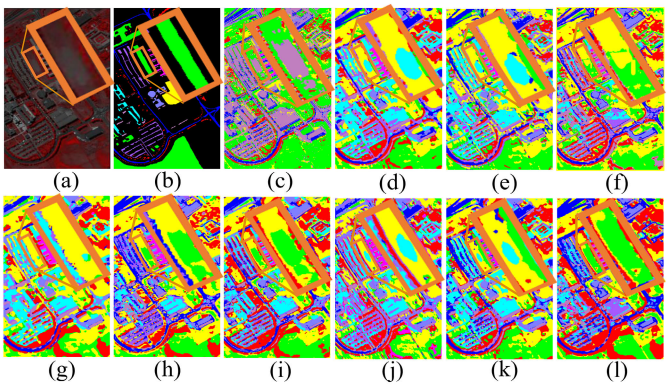



Fig. 11. Data visualization and the classification maps for the PU data obtained with different methods, including (a) false-color image of RP, (b) ground truth, (c) WFCG, (d) HyperSINet, (e) HyperSIGMA, (f) CMFSL, (g) CTF-SSCL, (h) Gia-CFSL, (i) DCFSL, (j) ADAFSL, (k) FDFSL, and (l) ACFSL.

classes have extremely limited labeled samples, ACFSL still achieves relatively accurate classification.

For the RP dataset, Fig. 10 shows that ACFSL can effectively alleviate class mixing compared with multiple baselines, reducing scattered misclassified pixels and producing more coherent and clearer classification maps. Furthermore, Fig. 11 illustrates the full-scene classification results for the PU dataset. This whole-scene visualization enables the evaluation of the model's performance even outside the explicitly annotated regions, further demonstrating its robust generalization capability.

To more comprehensively analyze the effectiveness and limitations of the proposed method, we compare different approaches from the perspective of computational complexity. Here, we mainly focus on FSL-based methods. As shown in Table VI, across the NW, TC, RP, and PU datasets, ACFSL consistently requires much less testing time than FDFSL. For example, on the TC dataset, ACFSL takes only 12.68 s for testing, whereas FDFSL requires 53.78 s. In terms of model size and computation, ACFSL has approximately 0.084M parameters and 0.39G–0.48G FLOPs. This is substantially smaller than CMFSL (0.64M parameters) and vastly more efficient than the resource-heavy Gia-CFSL (5.6G FLOPs), demonstrating that introducing synergistic global alignment and prototype calibration does not incur excessive computational overhead. While ACFSL trains faster than DCFSL on the NW and TC datasets, it requires more time on RP and PU to ensure sufficient global alignment and feature modeling, yet it remains within an acceptable range.

### D. Parameter Sensitivity Analysis

*a) Comparison of different numbers of annotated samples:* For each of the four TD datasets, we randomly select 1, 2, 3, 4, and 5 labeled samples per class to evaluate ACFSL and competing methods.

The results are shown in Fig. 12, where the shaded areas indicate the standard deviation. It can be observed that ACFSL consistently achieves the best performance under all labeling configurations, and its accuracy steadily improves as the number of labeled samples per class increases. As more labeled samples become available, the shaded regions of all methods gradually narrow, indicating improved stability. In contrast, supervised learning methods are clearly constrained in the few-shot setting. It is worth noting that even with only one labeled sample per class, ACFSL still achieves relatively high accuracy (e.g., 60.43% on NW and 62.88% on RP), which indicates that the proposed method can effectively transfer knowledge from SD to TD, thereby mitigating the adverse effects of domain shift.

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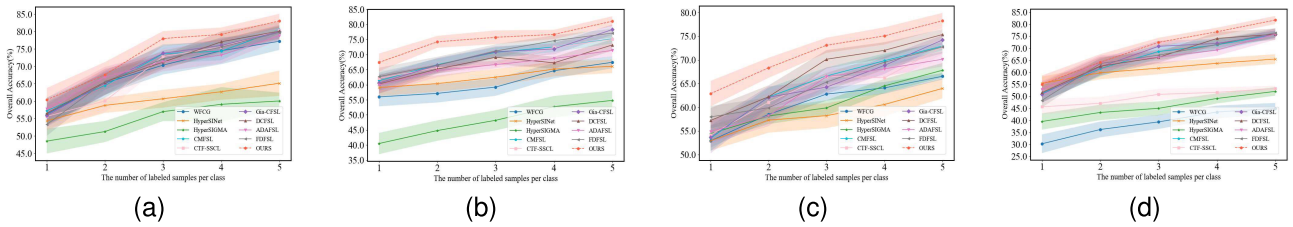



Fig. 12. OA (%) of different algorithms across four TD datasets with varying numbers of annotated samples. (a) NW. (b) TC. (c) RP. (d) PU.

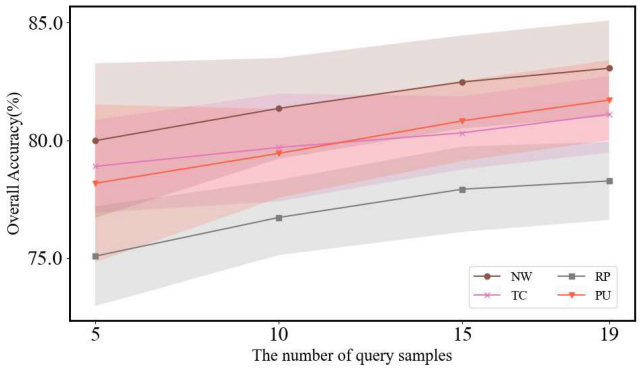



Fig. 13. OA (%) of ACFSL across four TD datasets with varying numbers of query samples.

*b) Comparison of different numbers of query samples:* We varied the number of query samples $N_q \in \{5, 10, 15, 19\}$ during the meta-training phase. The experimental results are presented in Fig. 13. The OA exhibits a downward trend as the query set shrinks, dropping from 81.10% ($N_q = 19$) to 78.89% ($N_q = 5$) on the TC dataset. However, even in the extremely constrained setting of $N_q = 5$, the OA only degrades by 2.21%. Notably, the resulting accuracy still surpasses the performance of Gia-CFSL (78.30%), even when the latter utilizes the full query set of $N_q = 19$. This confirms that the DPIC module remains highly effective without relying on a large query set. Because DPIC selectively aggregates only the top-tier typical samples and bottom-tier hard samples based on the relative prediction margins $m(z_q)$, it filters out uninformative samples, ensuring stable prototype updates and preventing prototype drift, even under small sample sizes.

*c) Comparison with different $\zeta_1$ and $\zeta_2$:* $\zeta_1$ and $\zeta_2$ are two key hyperparameters in DPIC, controlling the numbers of typical samples and hard samples involved in prototype calibration, respectively. For $\zeta_1$, an overly small value may be insufficient to capture representative intraclass patterns, whereas an overly large value may introduce noisy samples. We thus set $\zeta_1$ to range from 3 to 11 with a step size of 2, i.e., $[3, 5, 7, 9, 11]$. $\zeta_2$ directly influences the model's ability to exploit boundary and outlier samples. When $\zeta_2$ is set to a small value, the model may fail to mine discriminative cues from interclass confusing regions. Conversely, the prototype may become overly biased toward a small subset of hard or even noisy samples. Accordingly, we set $\zeta_2$ to $[3, 4, 5, 6, 7]$ to sufficiently explore the effect of hard-sample involvement on classification performance while maintaining prototype stability.

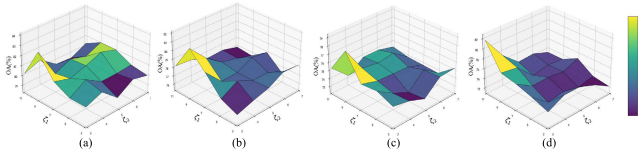



Fig. 14. Impact of different combinations of $\zeta_1$ and $\zeta_2$ on the OA (%). (a) NW. (b) TC. (c) RP. (d) PU.

Fig. 14 reports the OA results on the NW, TC, RP, and PU datasets under different parameter combinations. The results indicate that using a moderate number of typical samples ($\zeta_1 \approx 9$) helps form more stable and more representative class centers. Meanwhile, the incorporation of hard samples should be controlled; otherwise, it can easily introduce bias into the prototypes. When $\zeta_1$ is small (e.g., 3 or 5), OA is generally lower and more sensitive to variations in $\zeta_2$. As $\zeta_1$ increases, OA improves noticeably on all four datasets. However, when $\zeta_1$ further increases to 11, performance generally drops (e.g., on NW, the overall OA at $\zeta_1 = 11$ is lower than that at $\zeta_1 = 9$). Overall, setting $\zeta_2 = 3$ achieves the best or near-best performance, implying that hard samples do carry boundary information but tend to lie closer to interclass confusing regions; if too many are included, the hard sample prototype can be dominated by a small number of mixed or even outlier samples. Therefore, hard samples are more suitable as a small but precise correction term for decision boundaries rather than a dominant source of prototypes.

*d) Robustness to label flipping noise:* To further verify the robustness of the model against erroneous predictions in the transductive inference process, we corrupt the prediction results of query samples. Specifically, after the model generates initial predictions on the query set, we randomly select a certain ratio of query samples and flip their predicted class labels to other classes while retaining their original prediction confidence, so that these erroneously predicted samples are still treated as high-confidence typical samples and participate in the auxiliary prototype construction. The label flipping ratios are set to 0%, 10%, 20%, and 30%, respectively. The classification results are summarized in Table VII.

The OA exhibits a manageable degradation; even under a severe flipping ratio of 30%, the OA decreases by 2.85% on the NW dataset and 3.25% on the PU dataset. This robustness is attributed to the margin-based sample partitioning strategy and the bounded interpolation parameter $\alpha_{\max}$. The margin-based sorting ensures that only samples with stable prediction margins are selected as typical samples, while the bounded interpolation parameter restricts the maximum update

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TABLE VII

OA (%) OF ACFSL UNDER VARYING LABEL FLIPPING NOISE RATIOS ON THE FOUR TD DATASETS

| Dataset | | $P_{err}$=0% | $P_{err}$=10% | $P_{err}$=20% | $P_{err}$=30% |
|---|---|---|---|---|---|
| NW | OA(%) | 83.05±4.12 | 82.25±3.10 | 81.50±4.35 | 80.20±4.85 |
| | AA(%) | 85.62±1.28 | 84.37±2.95 | 83.80±3.01 | 82.55±2.75 |
| | Kappa (*100) | 77.83±4.17 | 77.05±4.58 | 76.85±5.20 | 75.45±5.65 |
| TC | OA(%) | 81.10±2.65 | 80.21±2.04 | 79.06±2.69 | 77.58±2.88 |
| | AA(%) | 78.92±1.85 | 78.10±2.71 | 77.14±1.94 | 75.65±2.17 |
| | Kappa (*100) | 76.25±2.80 | 75.13±2.66 | 73.96±2.90 | 72.43±3.13 |
| RP | OA(%) | 78.27±2.74 | 77.32±3.68 | 76.20±2.93 | 74.55±3.16 |
| | AA(%) | 78.72±3.48 | 77.49±3.92 | 76.46±3.69 | 75.85±3.96 |
| | Kappa (*100) | 73.14±2.96 | 73.68±2.90 | 72.36±3.15 | 71.29±4.43 |
| PU | OA(%) | 81.70±2.88 | 80.95±2.95 | 79.90±2.25 | 78.45±4.08 |
| | AA(%) | 78.56±2.56 | 77.85±2.70 | 76.20±3.51 | 74.75±4.65 |
| | Kappa (*100) | 76.61±2.12 | 75.85±2.52 | 74.60±2.06 | 72.95±5.10 |

TABLE VIII

CLASSIFICATION RESULTS OF ACFSL ON THE PU DATASET UNDER DIFFERENT CLASS IRS

| Class | $\gamma$=1.0 | $\gamma$=0.5 | $\gamma$=0.2 |
|---|---|---|---|
| 1(Majority) | 80.02±1.91 | 79.10±1.05 | 78.50±2.30 |
| 2(Majority) | 88.37±3.41 | 88.15±3.50 | 86.20±3.08 |
| 3(Minority) | 54.88±10.01 | 52.85±10.20 | 50.60±10.85 |
| 4(Majority) | 92.61±1.57 | 92.50±1.80 | 90.85±4.10 |
| 5(Majority) | 97.26±0.47 | 97.10±0.60 | 96.65±0.85 |
| 6(Minority) | 67.23±9.62 | 64.95±9.85 | 62.75±10.50 |
| 7(Minority) | 62.18±4.04 | 58.90±4.50 | 54.50±5.20 |
| 8(Minority) | 69.54±11.37 | 66.85±11.09 | 64.40±12.50 |
| 9(Majority) | 95.02±0.13 | 95.90±0.52 | 93.50±0.40 |
| OA(%) | 81.70±2.88 | 79.55±2.10 | 78.15±2.45 |
| AA(%) | 78.56±2.56 | 77.37±2.85 | 75.33±3.20 |
| Kappa (*100) | 76.61±2.12 | 75.30±2.30 | 73.45±2.65 |

amplitude of prototypes, preventing erroneous high-confidence predictions from dominating the prototype calibration process.

*e) Robustness to class imbalance:* To evaluate the robustness of the proposed model under class-imbalanced conditions, we conduct a sensitivity study on the PU dataset. We define the imbalance ratio (IR), defined as $N_{max}/N_{min}$, where $N_{max}$ and $N_{min}$ denote the numbers of query samples per majority and minority class, respectively. Five classes are randomly designated as majority classes ($N_{max}$ = 19), while the remaining four minority classes are downsampled to construct skewed query sets under IR = 2 and IR = 5. As summarized in Table VIII, even under IR = 5, where each minority class contains only approximately four query samples, the OA decreases by only 3.55% without severe performance degradation on minority classes. This is attributed to the per-class margin sorting mechanism of DPIC, where the class-level difficulty indicator $\bar{m}_k$ and the bounded interpolation parameter $\alpha_{max}$ are computed independently for each class; minority-class prototype calibration is effectively decoupled from the sample abundance of majority classes.

*f) Parameter analysis for SGDA:* The DA process involves several hyperparameters, including the MMD weight $\lambda_{mmd}$, the spatial-alignment weight $\lambda_{spa}$, and the spectral-alignment weight $\lambda_{spec}$. In contrast, the weight $\lambda_{adv}(t)$ is smoothly increased from 0 as training progresses.

To further investigate the relative importance of the loss weights, as shown in Fig. 15, we vary $\lambda_{mmd}$ from 0.1 to 1.0 and report the OA on the four TD datasets. Overall, all TD datasets achieve the best or near-best performance at relatively small $\lambda_{mmd}$ values, and OA exhibits a clear downward trend as $\lambda_{mmd}$ increases. Specifically, NW achieves the highest OA 83.05% at $\lambda_{mmd}$ = 0.1 and then gradually decreases, when $\lambda_{mmd}$ increases to 1.0, OA drops to 78.98%, by about 4.07%. This suggests that an overly large $\lambda_{mmd}$ imposes excessively strong domain-consistency constraints at the marginal-distribution level. We fix $\lambda_{mmd}$ = 0.1 in the subsequent experiments.

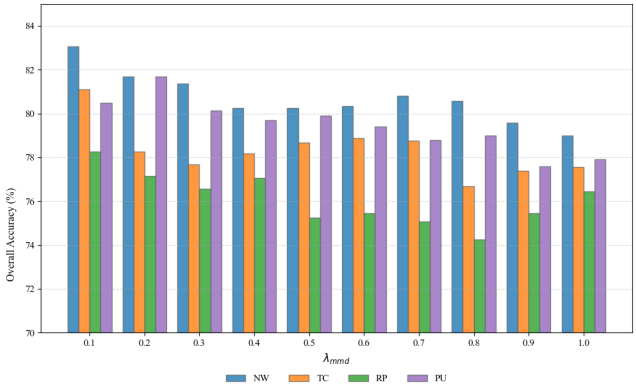



Fig. 15. OA with different coefficients of the MMD loss for the TD datasets.

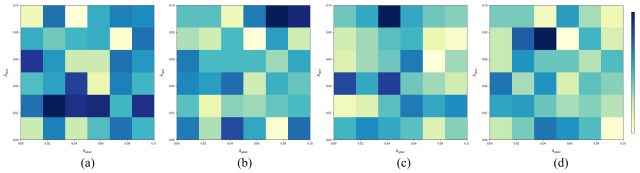



Fig. 16. OA with different coefficients of the spatial and spectral alignment losses for the TD datasets. (a) NW. (b) TC. (c) RP. (d) PU.

The experimental results for $\lambda_{spa}$ and $\lambda_{spec}$ are shown in Fig. 16. The overall trend suggests that both spatial and spectral alignment should be applied with moderate strength to achieve the best performance. On the NW dataset, OA reaches its peak when $\lambda_{spec}$ = 0.02 and $\lambda_{spa}$ = 0.1. However, when $\lambda_{spec}$ increases to 0.10, the performance exhibits noticeable fluctuations under different $\lambda_{spa}$ values, indicating that overly strong spectral distribution matching may blur crucial spectral differences among vegetation types. On the PU dataset, the best OA is achieved at $(\lambda_{spec}, \lambda_{spa})$ = (0.04, 0.08), suggesting that spatial alignment is better suited as a fine-grained structural constraint to preserve and strengthen discriminative texture patterns.

Overall, the loss weights exhibit broadly consistent trends across the four TD datasets, indicating that the proposed alignment structure generalizes well to different scenarios. Nevertheless, subtle performance differences can still be observed among datasets. Such differences mainly stem from the inherent wetland heterogeneity.

## E. Ablation Study and Visualization

This section presents comprehensive ablation studies and visual analyses to validate the rationale and individual contributions of the core modules within the ACFSL framework.

*a) t-SNE visualization of ACFSL:* To intuitively assess the effectiveness of ACFSL in feature extraction, we visualize both the raw spectral features and the network-extracted embedding

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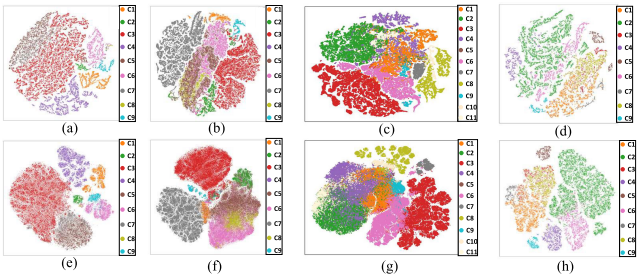


Fig. 17. t-SNE visualization of the original data and embedding features on the four datasets. (a) Original data of NW. (b) Original data of TC. (c) Original data of RP. (d) Original data of PU. (e) Embedding features of NW. (f) Embedding features of TC. (g) Embedding features of RP. (h) Embedding features of PU.

TABLE IX

COMPARISON OF CLASSIFICATION PERFORMANCE BETWEEN INDUCTIVE AND TRANSDUCTIVE SETTINGS

| Dataset | Inductive | Transductive | OA(%) | AA(%) | Kappa (*100) |
|---|---|---|---|---|---|
| NW | ✓ | | 81.91±4.74 | 84.79±1.38 | 76.64±5.50 |
| | | ✓ | **83.05±4.12** | **85.62±1.28** | **77.83±4.17** |
| TC | ✓ | | 78.55±2.07 | 76.90±2.13 | 74.78±3.49 |
| | | ✓ | **81.10±2.65** | **78.92±1.85** | **76.25±2.80** |
| RP | ✓ | | 75.79±3.25 | 77.24±3.85 | 72.19±4.12 |
| | | ✓ | **78.27±2.74** | **78.72±3.48** | **73.14±2.96** |
| PU | ✓ | | 78.90±2.75 | 76.84±3.24 | 74.10±2.97 |
| | | ✓ | **81.70±2.88** | **78.56±2.56** | **76.61±2.12** |

features of the four TD datasets using T-distributed stochastic neighbor embedding (t-SNE), as shown in Fig. 17. Evidently, in the raw spectral feature space, the TD samples are widely scattered, and different classes exhibit substantial overlap. For example, in Fig. 17(a), samples from classes 3, 5, and 7 in the NW dataset are heavily mixed in the 2-D space, and classes 5, 6, and 8 are difficult to separate with clear boundaries of TC; in the RP dataset, classes 1, 4, and 11 show an apparent overlap trend. In Fig. 17(d), the raw PU dataset exhibits this, particularly among classes 2, 4, and 6, which entirely lack clear decision boundaries.

The embedding features encoded by the ACFSL feature extractor [Fig. 17(d)–(f)] exhibit a noticeable structural change, showing improved intraclass compactness and interclass separability in the embedding space. For example, in the NW dataset, the previously highly confused classes 3, 5, and 7 become clearly separable. In the RP dataset, the boundary between classes 1 and 11 is also more distinct, forming more well-defined cluster structures. On the TC dataset, intraclass clustering is substantially improved as well, although a certain degree of overlap remains between some classes. Furthermore, on the PU dataset, the heavily entangled classes 2, 4, and 6 are successfully disentangled into highly cohesive and distinct clusters.

*b) Ablation study of DPIC:* To further validate the effectiveness of the DPIC module, we compare the proposed transductive ACFSL with its pure inductive variant. For the inductive variant, we completely disabled the joint query processing mechanism. As shown in Table IX, the transductive ACFSL consistently outperforms its inductive variant, yielding OA gains of 1.14%, 2.55%, 2.48%, and 2.80% on NW, TC, RP, and PU, respectively. This verifies that the prototype calibration guided by prediction confidence is both reasonable and effective. Meanwhile, the inductive variant alone still achieves competitive accuracy, confirming that the overall superiority of ACFSL does not stem exclusively from the transductive evaluation protocol. The robust spectral–spatial representations extracted by CPFE and the distribution shift mitigation enforced by SGDA also play a pivotal role in maintaining generalization.

TABLE X

ABLATION STUDY ON SGDA

| | $\lambda_{spa}$ | $\lambda_{spec}$ | NW | TC | RP | PU |
|---|---|---|---|---|---|---|
| **Baseline** | 0.1 | 0.06 | **82.51** | **80.76** | **77.83** | **80.85** |
| **w/o All** | 0 | 0 | 79.02 | 78.45 | 75.68 | 78.22 |
| **w/o Spa** | 0 | 0.06 | 79.54 | 78.79 | 76.04 | 78.63 |
| **w/o Spec** | 0.1 | 0 | 80.16 | 79.20 | 76.32 | 78.94 |

TABLE XI

ABLATION STUDY ON SPATIAL ALIGNMENT

| Dataset | w/o Spatial Alignment | +Adv Only | +MMD Only | +Adv and+MMD |
|---|---|---|---|---|
| **NW** | 79.67 | 80.95 | 82.30 | **83.05** |
| **TC** | 78.41 | 79.96 | 79.53 | **84.68** |
| **RP** | 75.44 | 76.78 | 76.21 | **79.87** |
| **PU** | 78.52 | 79.02 | 79.35 | **81.70** |

*c) Ablation study of SGDA:* To evaluate the contribution of each alignment component to the overall performance, we use the full ACFSL model as the baseline. The loss-weight configuration is set as follows: the adversarial loss adopts the aforementioned dynamic scheduling strategy, with $\lambda_{spa} = 0.1$ and $\lambda_{spec} = 0.06$. Based on this baseline, we then remove individual alignment terms by setting the corresponding weight to zero and record the resulting changes in TD performance.

As reported in Table X, the complete SGDA framework achieves the best performance across four TD datasets. Removing all alignment terms leads to a clear degradation; on the NW dataset, OA drops from 82.51% to 79.02%. Moreover, ablating any single component consistently reduces classification accuracy; for the PU dataset, removing the spectral alignment module causes the OA to decrease by 1.91%. Furthermore, Table XI investigates the internal synergy of the spatial alignment module's Adv and MMD subcomponents. While applying either constraint individually yields only marginal gains, integrating both simultaneously provides a substantial performance boost across all datasets. For instance, their combination increases the OA on the PU dataset from 78.52% to 81.70%, demonstrating their strong collaborative effectiveness.

*d) Ablation study of CPFE:* To evaluate the individual contributions of the SSAD and DGFM components within the CPFE, an ablation study is conducted and the results are presented in Table XII.

The baseline model employs a standard convolutional backbone without these specialized feature calibration mechanisms. When the SSAD is integrated into the baseline independently, the OA improves steadily across all four datasets. For instance, the accuracy on the NW dataset increases from 80.46% to 81.52%. This improvement demonstrates that adaptively recalibrating informative spectral channels and salient spatial locations amplifies subtle interclass variations. Similarly,

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TABLE XII
ABLATION STUDY OF THE CPFE

| Dataset | Baseline | SSAD | DGFM | OA(%) |
|---|---|---|---|---|
| NW | ✓ | | | 80.46 |
| | ✓ | ✓ | | 81.52 |
| | ✓ | | ✓ | 81.75 |
| | ✓ | ✓ | ✓ | **83.05** |
| TC | ✓ | | | 79.79 |
| | ✓ | ✓ | | 80.24 |
| | ✓ | | ✓ | 80.37 |
| | ✓ | ✓ | ✓ | **81.10** |
| RP | ✓ | | | 75.93 |
| | ✓ | ✓ | | 76.54 |
| | ✓ | | ✓ | 76.69 |
| | ✓ | ✓ | ✓ | **78.27** |
| PU | ✓ | | | 78.32 |
| | ✓ | ✓ | | 79.45 |
| | ✓ | | ✓ | 79.79 |
| | ✓ | ✓ | ✓ | **81.70** |

TABLE XIII
ABLATION STUDY OF EACH MODULE IN ACFSL

| Dataset | Baseline | FSL | CPFE | DPIC | SGDA | OA(%) | AA(%) | Kappa (*100) |
|---|---|---|---|---|---|---|---|---|
| NW | ✓ | | | | | 63.89±5.14 | 66.94±7.42 | 60.11±5.45 |
| | ✓ | ✓ | | | | 75.63±5.89 | 76.58±2.65 | 69.38±8.79 |
| | ✓ | ✓ | ✓ | | | 80.17±4.49 | 83.34±1.30 | 74.62±4.83 |
| | ✓ | ✓ | ✓ | ✓ | | 81.25±5.87 | 84.27±2.13 | 75.23±6.14 |
| | ✓ | ✓ | ✓ | ✓ | ✓ | **83.05±4.12** | **85.62±1.28** | **77.83±4.17** |
| TC | ✓ | | | | | 63.89±4.52 | 60.48±5.92 | 56.89±6.46 |
| | ✓ | ✓ | | | | 73.93±4.04 | 69.91±2.64 | 67.76±1.90 |
| | ✓ | ✓ | ✓ | | | 78.66±3.55 | 77.61±1.50 | 74.37±3.48 |
| | ✓ | ✓ | ✓ | ✓ | | 79.74±3.20 | 78.09±1.79 | 75.38±2.40 |
| | ✓ | ✓ | ✓ | ✓ | ✓ | **81.10±2.65** | **78.92±1.85** | **76.25±2.80** |
| RP | ✓ | | | | | 60.65±4.31 | 61.24±2.59 | 52.99±2.98 |
| | ✓ | ✓ | | | | 70.37±3.92 | 68.83±4.16 | 62.61±3.56 |
| | ✓ | ✓ | ✓ | | | 75.91±2.04 | 77.76±2.07 | 69.04±2.03 |
| | ✓ | ✓ | ✓ | ✓ | | 76.34±1.43 | 78.03±2.85 | 71.47±3.26 |
| | ✓ | ✓ | ✓ | ✓ | ✓ | **78.27±2.74** | **78.72±3.48** | **73.14±2.96** |
| PU | ✓ | | | | | 64.72±3.20 | 62.84±1.13 | 57.26±1.96 |
| | ✓ | ✓ | | | | 74.10±3.75 | 69.27±2.04 | 68.10±2.74 |
| | ✓ | ✓ | ✓ | | | 77.98±3.27 | 75.70±1.84 | 73.24±2.63 |
| | ✓ | ✓ | ✓ | ✓ | | 79.05±3.93 | 75.82±2.14 | 74.30±3.06 |
| | ✓ | ✓ | ✓ | ✓ | ✓ | **81.70±2.88** | **78.56±2.56** | **76.61±2.12** |

solely adding the DGFM to the baseline yields notable performance gains. This gain confirms that coupling long-range dependency modeling with global spatial context enhances the representation of complex geomorphological structures. Furthermore, integrating both modules simultaneously leads to the highest classification accuracy. The complete feature extractor achieves an OA of 81.10% on the TC dataset. These results verify that the local feature recalibration provided by the SSAD and the global context fusion enabled by the DGFM operate synergistically to extract highly discriminative and robust spectral–spatial embeddings.

*e) Ablation study of ACFSL components:* To verify the effectiveness of each strategy in ACFSL, we design a comprehensive ablation study; the results are reported in Table XIII.

The first row presents the baseline performance by removing all modules, where the baseline refers to the classification performance obtained using only the backbone feature extraction network. As can be seen, this network yields the worst performance and lacks cross-dataset stability; e.g., its OA on the NW dataset is only 63.89%. We then deploy a variant that retains only the CDFSL framework. Since both SD and TD data are used during meta-training, the performance is improved over the baseline, with OA on the NW dataset increasing to 75.63%. However, this variant still exhibits limited performance. Next, we introduce the SGDA module. The results show a clear improvement, with OA on the TC dataset increasing from 73.93% to 78.66%. This gain confirms the critical role of the domain alignment module in mitigating distribution shifts. The fourth row of Table XIII further reports the results after incorporating DPIC, where consistent performance improvements are observed across all datasets, indicating that DPIC enhances prototype discriminability via the joint optimization of typical and hard samples. Finally, the fifth row presents the results of the full ACFSL model that integrates all core modules. It achieves the best performance on all test datasets, with OA reaching 83.05%, 81.10%, 78.27%, and 81.70% on the NW, TC, RP, and PU datasets, respectively. These results validate the synergistic effect among the proposed modules and demonstrate that each component contributes meaningfully to the overall performance.

## V. CONCLUSION

This article proposes a CDFSL method (ACFSL) via adaptive calibration for HSIC. By introducing SSAD and DGFM to construct a CPFE, the method yields more compact and discriminative embeddings. Furthermore, to address the issue of inaccurate prototype estimation induced by sample scarcity and boundary ambiguity, we propose a DPIC strategy, which improves intraclass compactness and interclass separability. Finally, we design an SGDA strategy that aligns macroscopic spatial distributions and spectral baselines, effectively mitigating domain shift while preserving class-discriminative structures. Extensive comparative experiments and ablation studies on four datasets demonstrate that ACFSL consistently achieves leading performance in terms of classification accuracy and stability compared with representative state-of-the-art methods. However, the present study remains focused on single-date hyperspectral few-shot classification. It does not fully model broader ecological factors such as tidal dynamics, vegetation structural variation, or long-term wetland succession.

In future work, we will integrate multitemporal remote sensing data, hydrological information, and ecological prior knowledge to further improve the ecological interpretability and applicability of coastal wetland classification.

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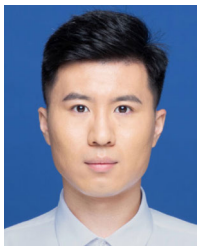
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